



ITI

Introduction to Computer Networks



Computer Network Workshop



References:

- Essential Computer Science “ Paul D. Crutcher, Neeraj Kumar Singh, Peter Tiegs”
- <https://maharatech.gov.eg/course/view.php?id=2116>
- Cisco Student Guide ICND1
- CompTIA Network



Session 1 (Network Essentials)



- **Session Outlines**

- **Computer Networks**

- Definition and Basic Terminologies

- **TCP/IP Protocol Suite**

- Network Access Layer (Physical Layer /Datalink Layer)
 - Internet layer
 - Transport layer
 - Application Layer



Session 1 (Computer Networks Definition)

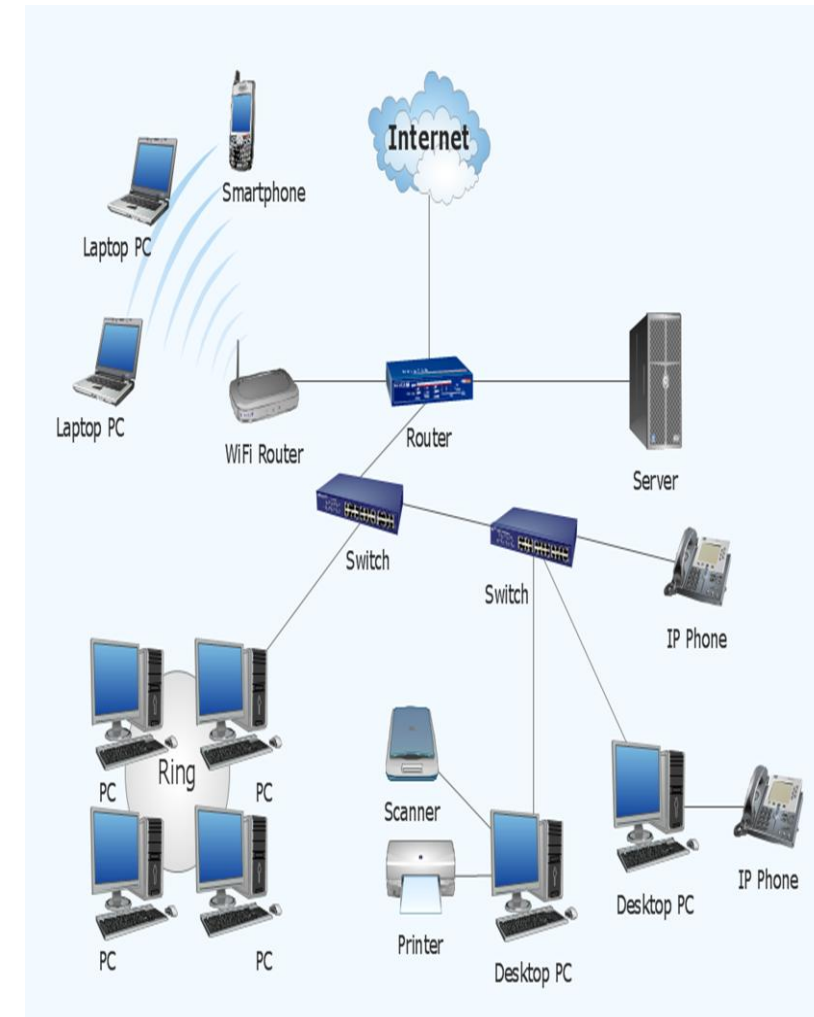
- **Computer Network :**

- a collection of computers, and other devices, or peripherals connected together through connecting media to perform certain task such as :

Share Resources

- **Resources can be :**

- File Sharing
- Devices Sharing
- Software Sharing with multi-user licenses.
- Voice and Video calls
- Shared Internet Access



Session 1 (Network Elements)



- **Network Elements**
Hardware

- **Devices**

- Computers – Printers –Phone – Routers - Switches

- **Medium**

- Wired -Wireless –Satellites

- **Software**

- **Messages**

- Information that travels over the medium such as Mails-WhatsApp....etc

- **Protocols**

- Governs how messages flow across network such as http –https-FTP-RDP



Session 1 (Network Basic Terminologies)



- **NIC (Network Interface Card)/network adapter or LAN adapter.**
 - a hardware that enable the device to directly access the network
 - Internal NIC (plugs into the motherboard directly)
 - External NIC (Wireless and USB based)
- **Mac address:**
 - Physical Address, Unique address over the world burned on the NIC card
- **IP address :**
 - logical address, identify each device on an IP network layer.
- **Protocols**
 - Communication rules that all entity must agree on http –https-FTP-RDP



Session 1 (Network Basic Terminologies)



- **Hub**

- Allow different nodes to communicate with each other at the same network(Slow the network)



- **Switch**

- Allow different nodes to communicate with each other at the same network and time without slowing each other



- **Repeater**

- Regenerate the signal over the same network before the signal becomes too weak or corrupted



- **Access point (AP)**

- allows other Wi-Fi devices to connect to a wired network. An AP is a physical location where Wi-Fi access is available.



- **Router**

- Allow different networks to communicate with each other



How to apply networks ?



- **According to Covered Area**
 - How large is the network?
- **According to network topology**
 - How the computer are connected?
- **According to network model**
 - What type of model?



Networks Classifications



According to Covered Area

PAN – MAN- WAN-INTERNET



According to Covered Area

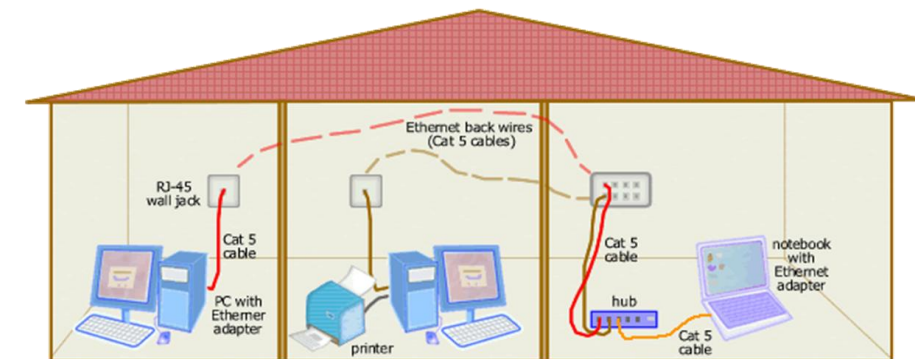
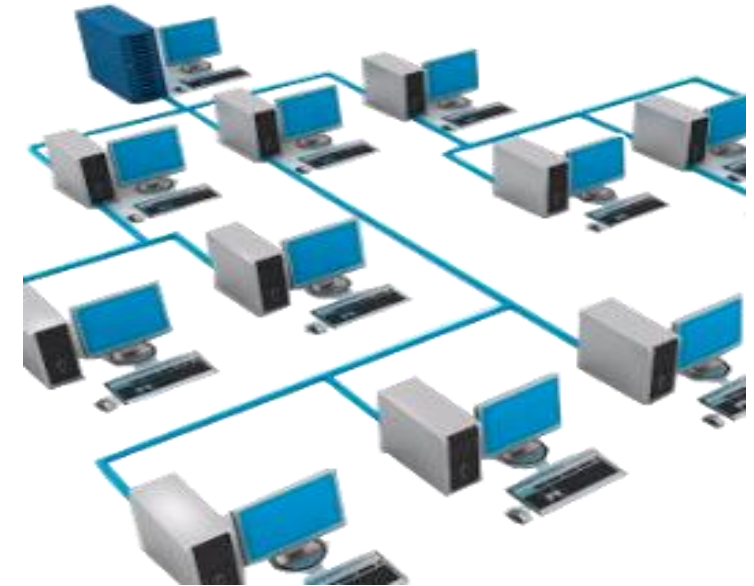
- Personal Area Networks (**PAN**)
 - a computer network for interconnecting devices centered on an **individual person's workspace**.
 - A **PAN** provides data transmission among devices such as computers, smartphones, tablets and personal digital assistants



According to Covered Area

- Local Area Networks (**LAN**)

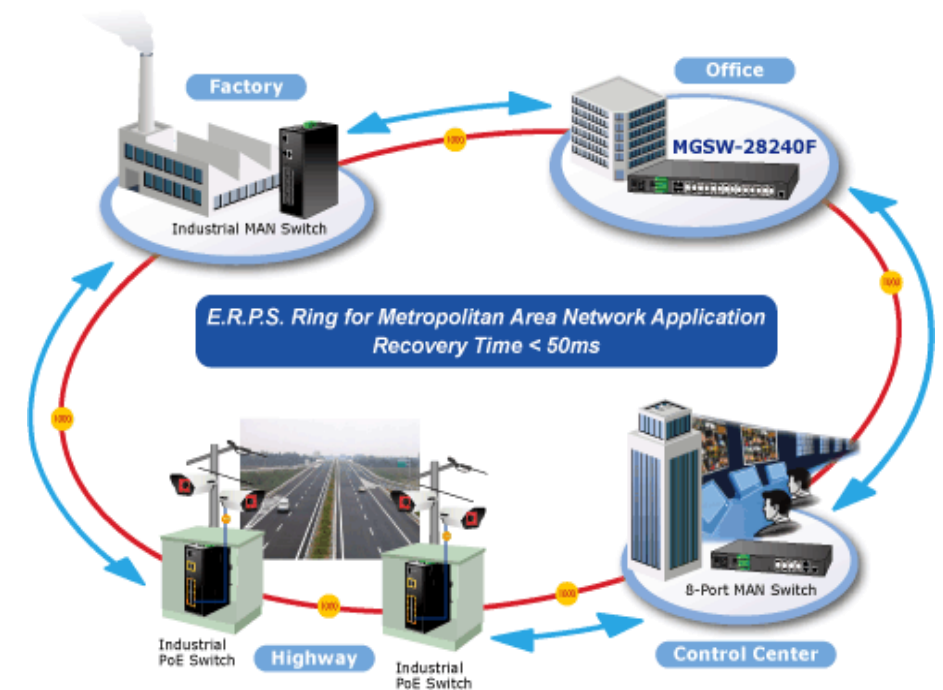
- a **group** of computers connected in **small** geographical area
- a limited area such as a residence, school, laboratory, university campus or office building (100 -1000 M)
- Allow users to share files and services
- **High speed** of communications
- Under your **administrative Control**



According to Covered Area



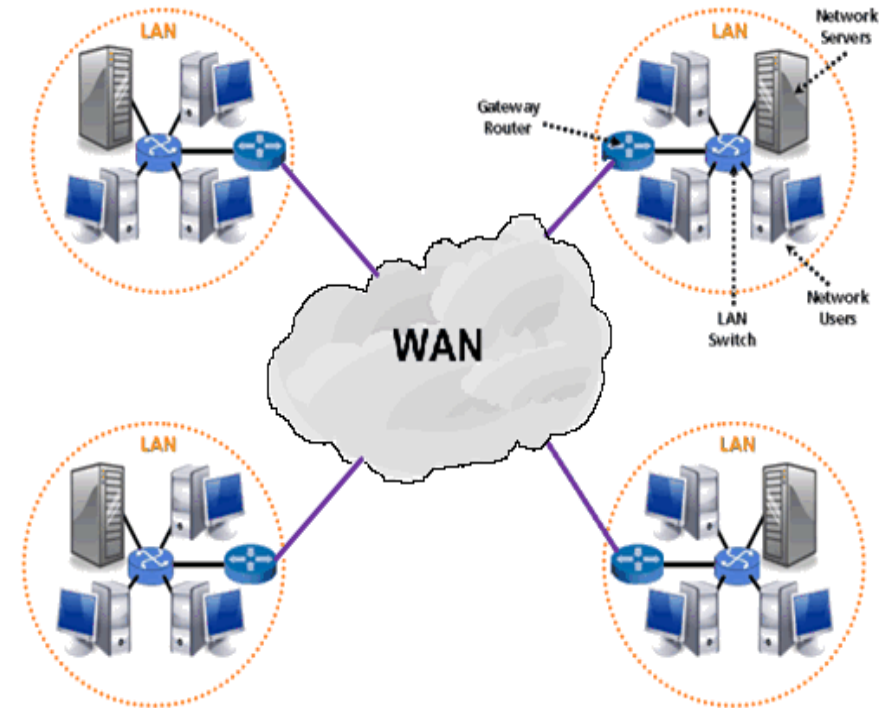
- Metropolitan Area Networks (**MAN**)
 - A MAN connects an **area larger than a LAN but smaller than a WAN** (Up to 100 km)
 - such as a **city**.
 - dedicated or high-performance hardware



According to Covered Area

- Wide Area Networks (**WAN**)

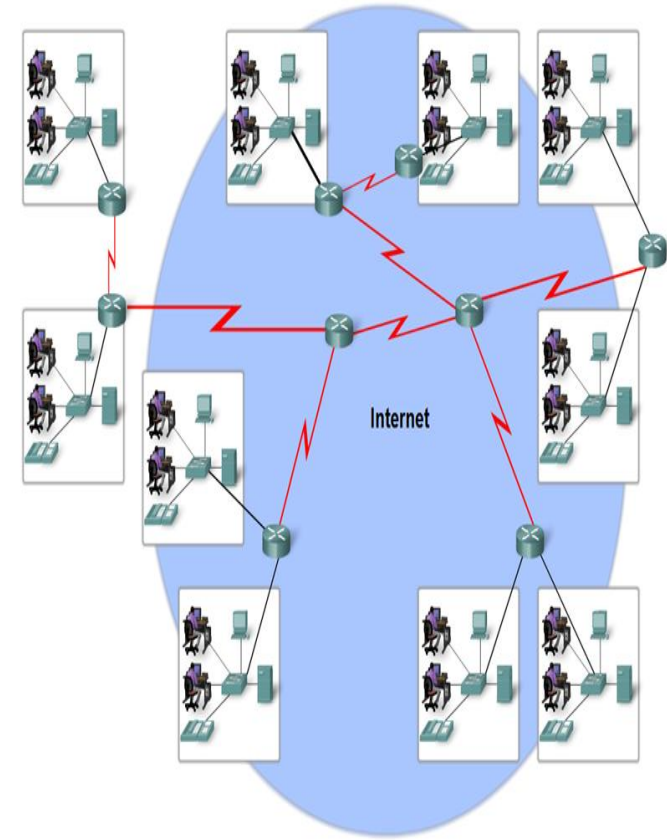
- A WAN is a group of computers connected in **Large geographical** area such as **country**
- A WAN often connects two LANs (WAN Link)
- WAN can contain multiple smaller networks, such as LANs or MANs.
- Very low Speed
- Under your **ISP** Administrative control
example of WAN is **Internet**



What is Internet ?

- **Internet (WWW)**

- The internet is defined as a **global mesh** of interconnected networks
- the most used service on the Internet is the **World Wide Web**
- No one actually owns the Internet
- Many Orgs, ISPs, Companies, Govs own pieces of Internet Infrastructure.
 - ISOC: Internet Society
 - IETF: Internet Engineering Task Forum
 - **ICANN**: Internet Corporation for Assigned **N**ames and **N**umbers



Networks Classifications



According to Network Topology

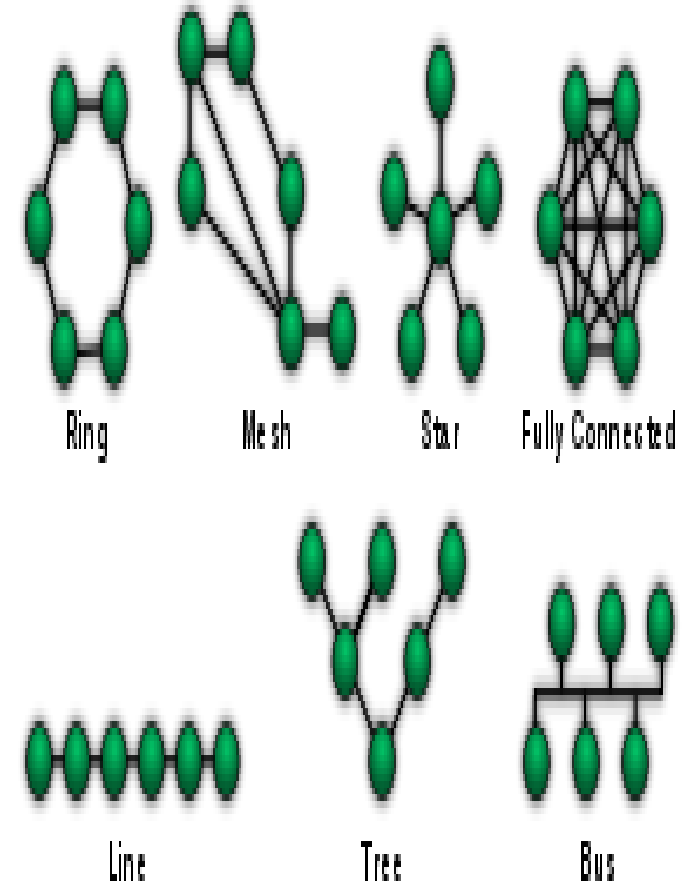
Bus – Star- Ring- Mesh-Hybrid

Network Topology

Topology

- Refers to the **shape of a network**, or the **network's layout**.
- **Types**
 - **Physical Topology:** how computers connected to each other physically (wired)
 - **Logical Topology:** how to send message from device to other, (the way in which to the generated signal actual path across the network).
- **Dependent on :**
 - **Type and number of equipment being used**
 - **Cost**

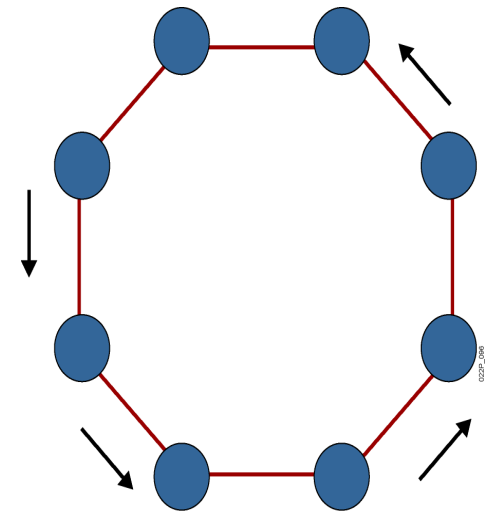
Each topology has its own **advantages and disadvantages**.



Network Topology

Ring Topology

- A cable connects one node to another to form a ring (shape of a closed loop)
- each device is connected directly to two other devices, one on either side of it.
- All messages travel through a ring in **the same direction**
- **token** is used to transmit data and pass over each station
- **Medium access control** determines when station can insert frame

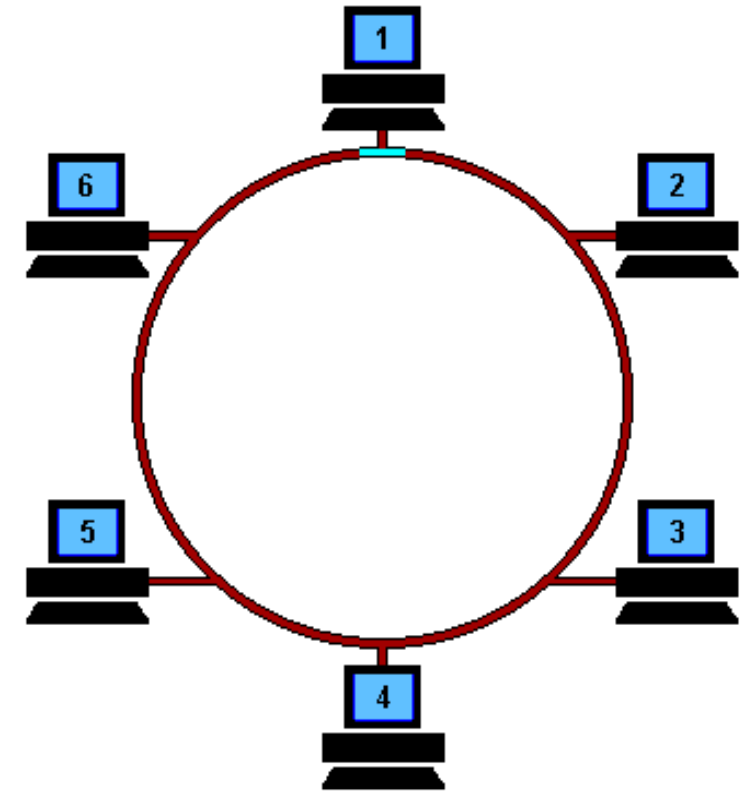


Network Topology

Frame Transmission - Ring LAN

Data transmitted in frames (token)

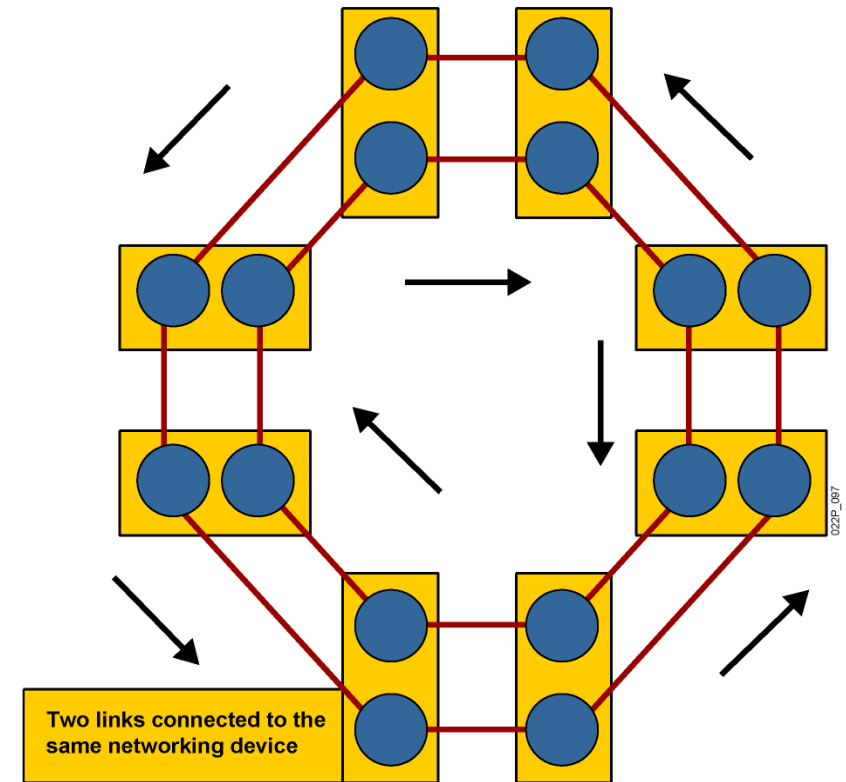
- ❑ Circulate past all stations
- ❑ Destination recognizes address and copies frame
- ❑ Data is passed one way from device to device.
- ❑ Frame circulates back to source where it is removed
- ❑ **Medium access control** determines when station can insert frame



Network Topology

Dual Ring Topology

- Signals travel in opposite directions.
- More resilient than single ring.



Network Topology

Ring Advantages and Disadvantages

❑ Advantages:

- **Fair** (Equal access for all users)
- Perform **well** under heavy traffic

❑ Disadvantages

- Network expansion or reconfiguration will affect the network operation
- If **one node** fails, the entire network fails
- Difficult to troubleshoot
- very bad if we have about **60** pc, Slow Network



Network Topology

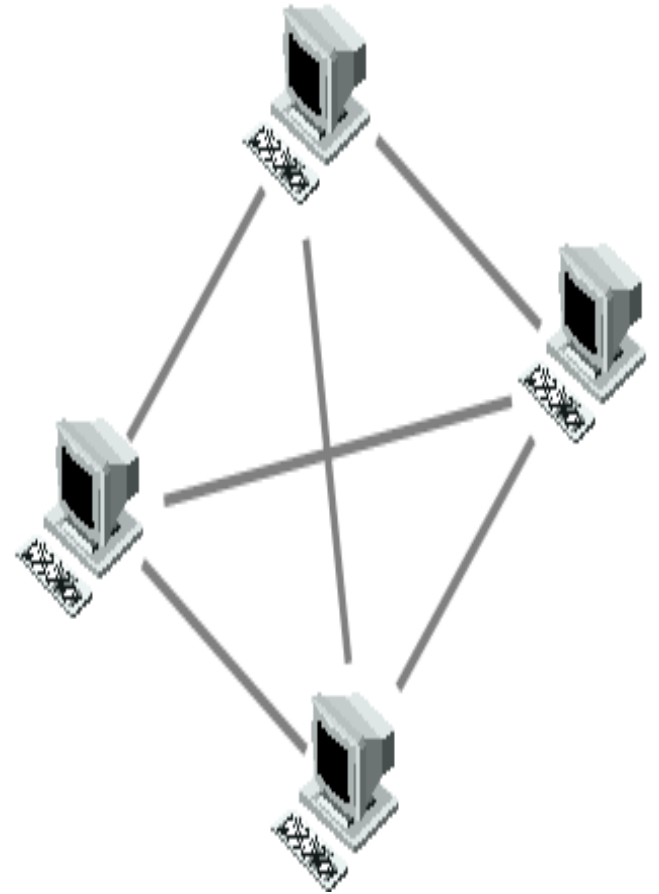
Mesh Topology

- Each device/PC is connected to **every** other device/PC in the network by its own cable
- Amount of cables needed can be calculated by

$$CN = (D * (D-1)) / 2$$

(where CN is Cables Needed, and D is the amount of devices on the network)

- Mesh Types:
 - Full Mesh
 - Partial Mesh



Network Topology



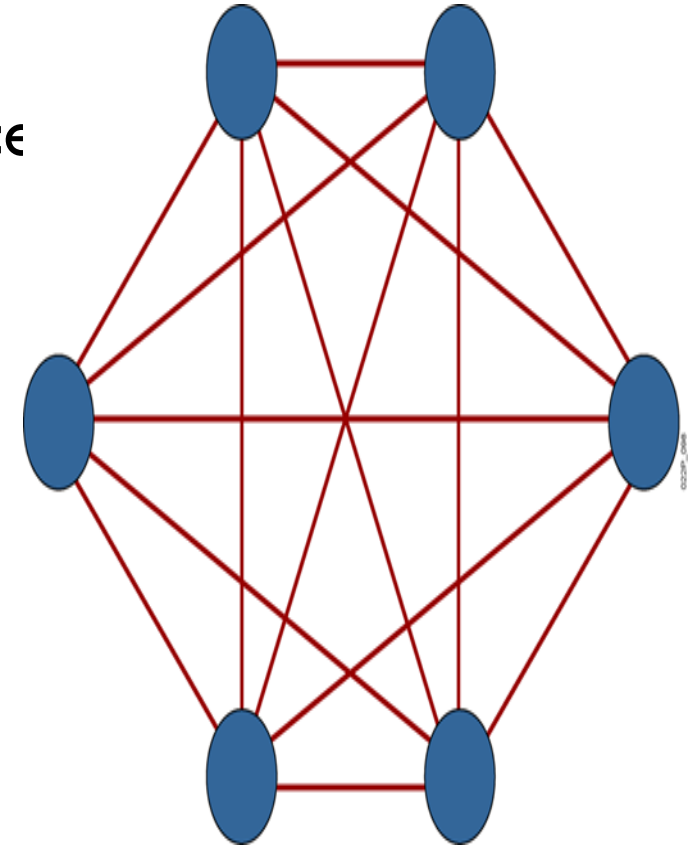
Mesh Advantages and Disadvantages

□ Advantages:

- Mesh topology boasts the highest fault tolerance all of the network topologies
- **Redundancy** exist
- **Secure**

□ Disadvantages

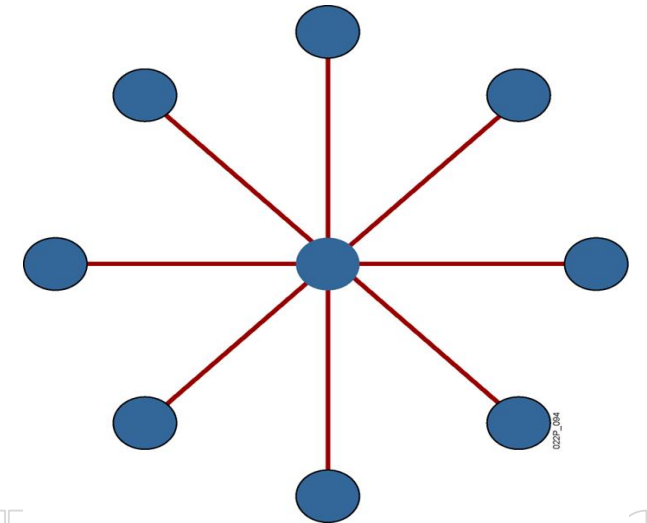
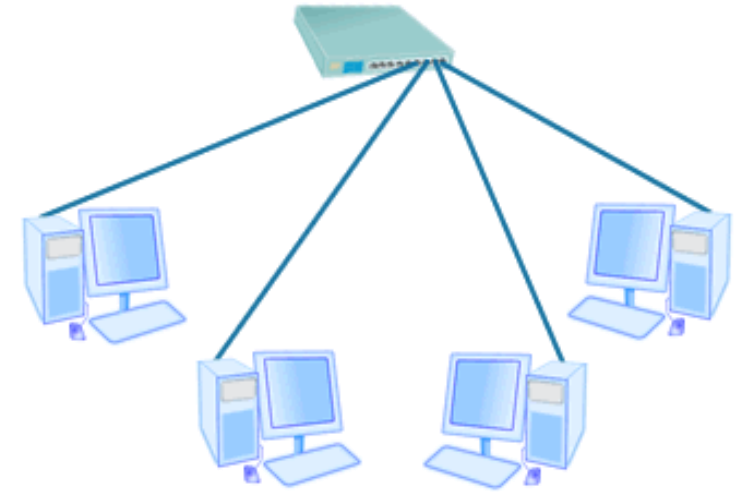
- Because each connection needs its own cable a Mesh topology can get **very expensive**



Network Topology

Star Topology

- All the devices are connected to a **centralized unit** such as a Hub or Switch.
- Nodes communicate across the network by passing data through the central device.

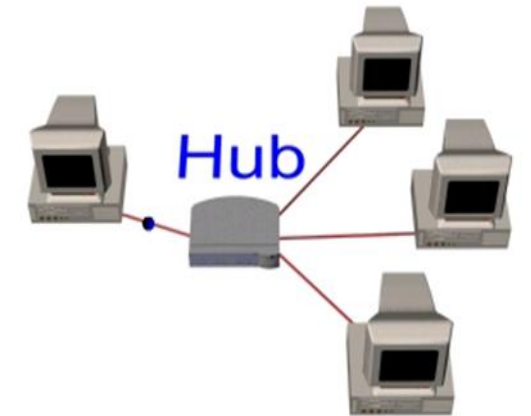


Network Topology

Star Topology types

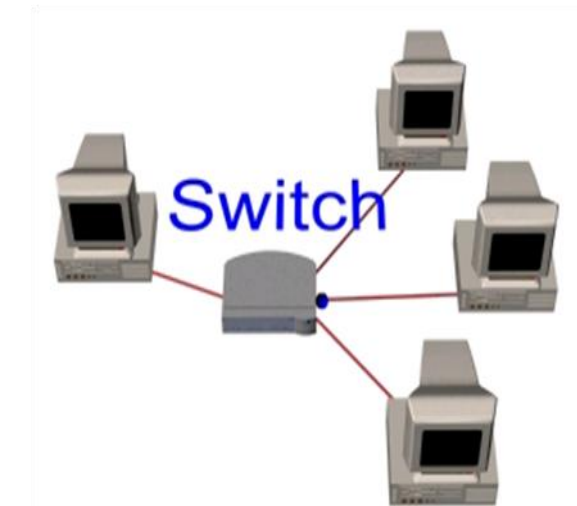
❑ Hubbed Star (Broadcasted Star Topology)

- Central node can broadcast (Hub)
 - Physical star, logically bus
 - **Only one station can transmit at a time**



❑ Switched Star

- Central node can act as frame switch
 - Retransmits only to destination



Network Topology

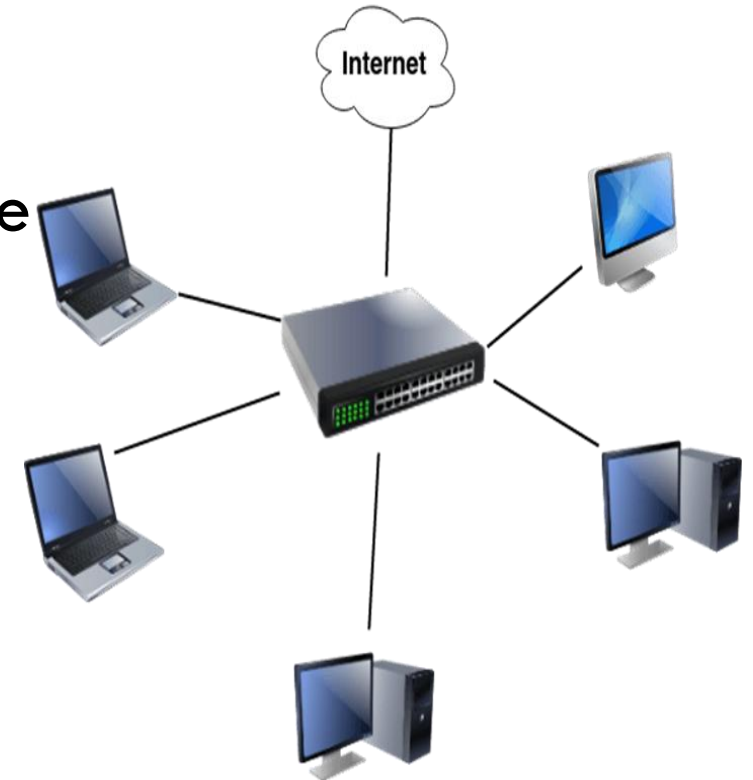
Star Advantages and Disadvantages

❑ Advantages:

- Network not affected if one PC fails
- Network expansion and reconfiguration is simple
- Network management and monitoring can be centralized
- Troubleshooting is easy

❑ Disadvantages

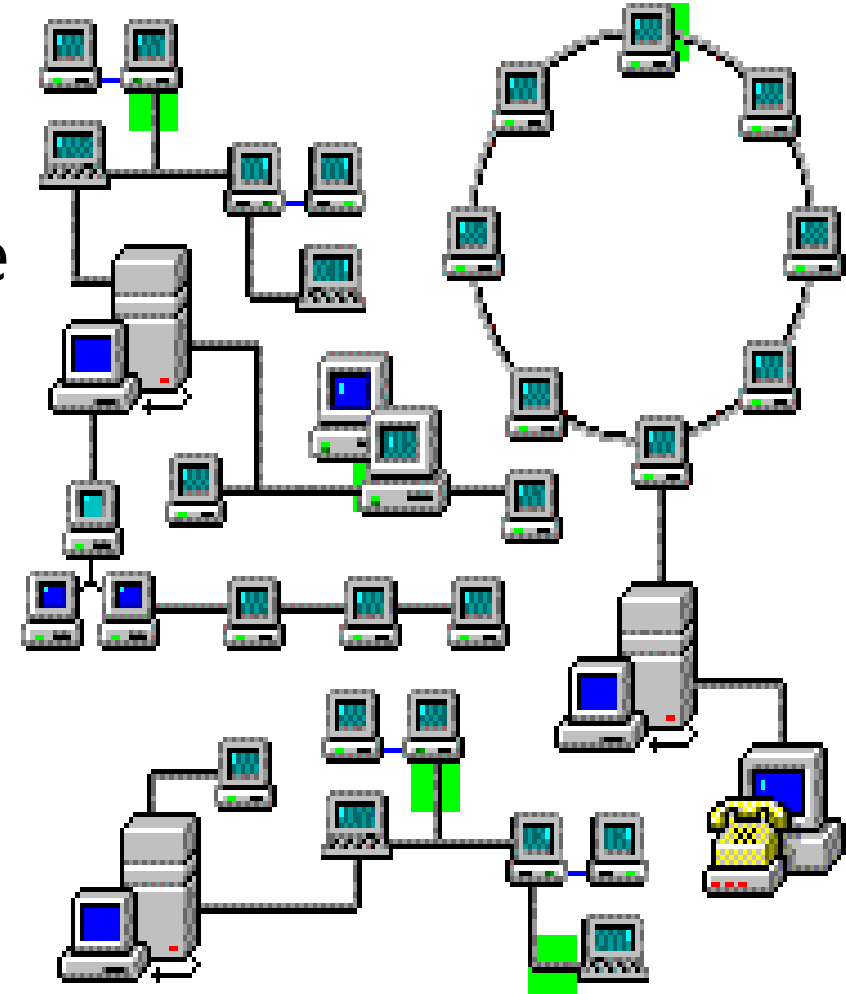
- If the central device fails, all the network fails



Network Topology

Hybrid Topology

- Hybrid means that there is more than one topology exist
- Combine bus ,star and ring topologies
- Allow network expansion
- Flexible



Networks Classifications



According to Network Model

- Peer to Peer Networks
- Client/Server Networks

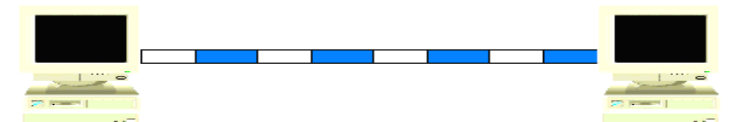
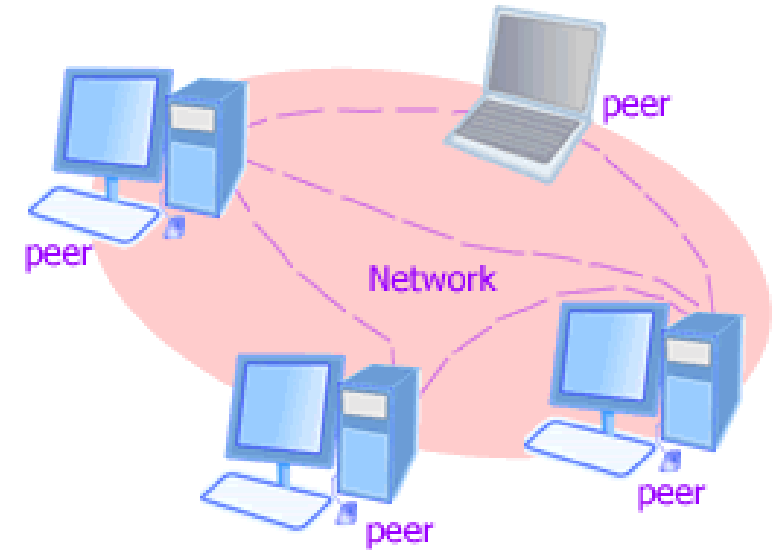


Network Model

Peer-to-Peer Networks

- **No** dedicated resources to present specific service
- **Easy** to work with
- **All nodes are the same** (equal to use the resources)

Example : Windows Workgroup



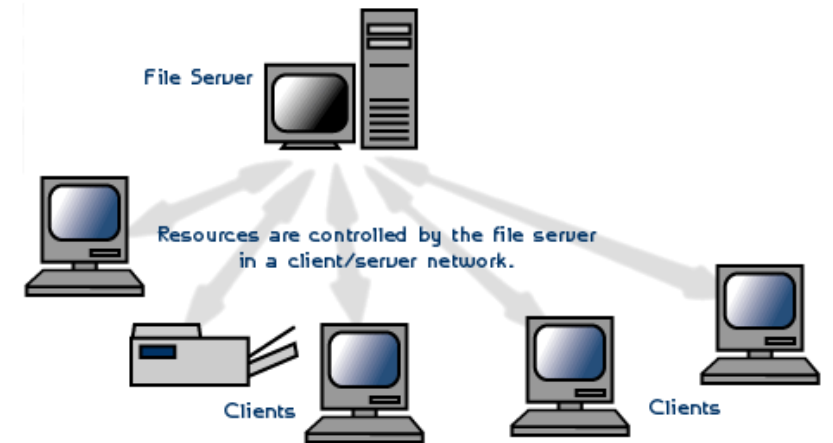
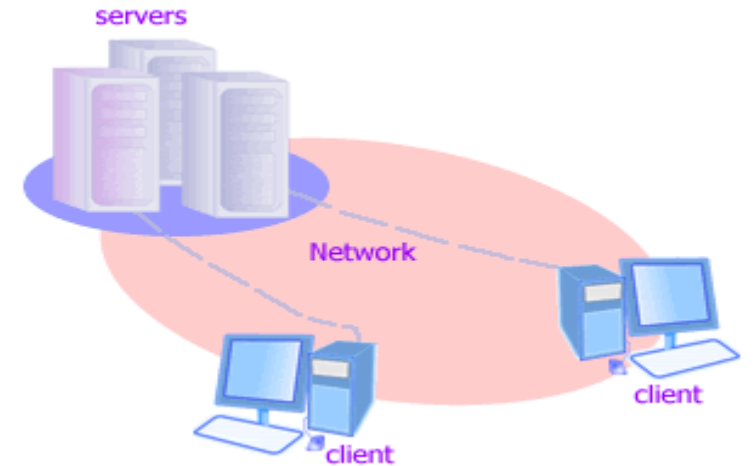
Network Model

Client/Server Networks

- Some nodes (**SERVER**) are dedicated to **present services** to other nodes (**CLIENTS**)
- Server is **more powerful**

Examples:

- Mail Server
- Web Server
- File Server
- Print Server

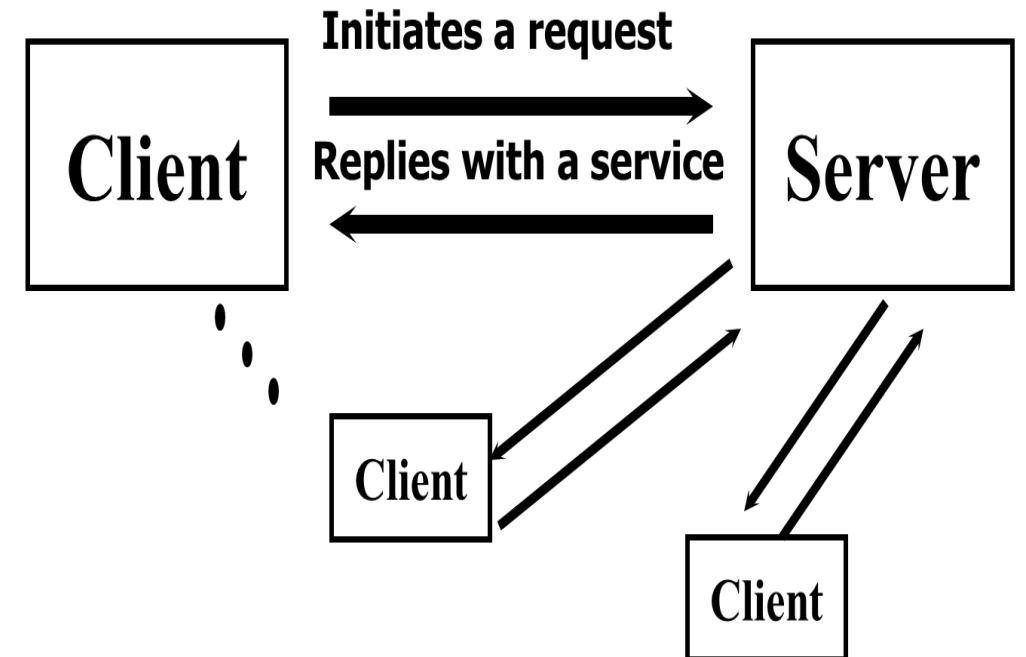


Network Model



Client/Server Networks

- computers providing the service are called **Servers**
- computers that request and use the service are called **Client** computers.
- number of servers is very small compared with the number of clients



Peer to peer vs client/server model



	Peer to peer	Client/server
Centralization	Local machine no central server	All client machines connect to central server to get service
Storage	Each machine share its files equally with the others	All files and folders are on dedicated storage on the server and client access their files based on database on the server
Cost	inexpensive	Expensive because of server OS license
scalable	In home or small office	Medium/large enterprise
Operating system	Client operating system	Server operating system to handle multiple requests



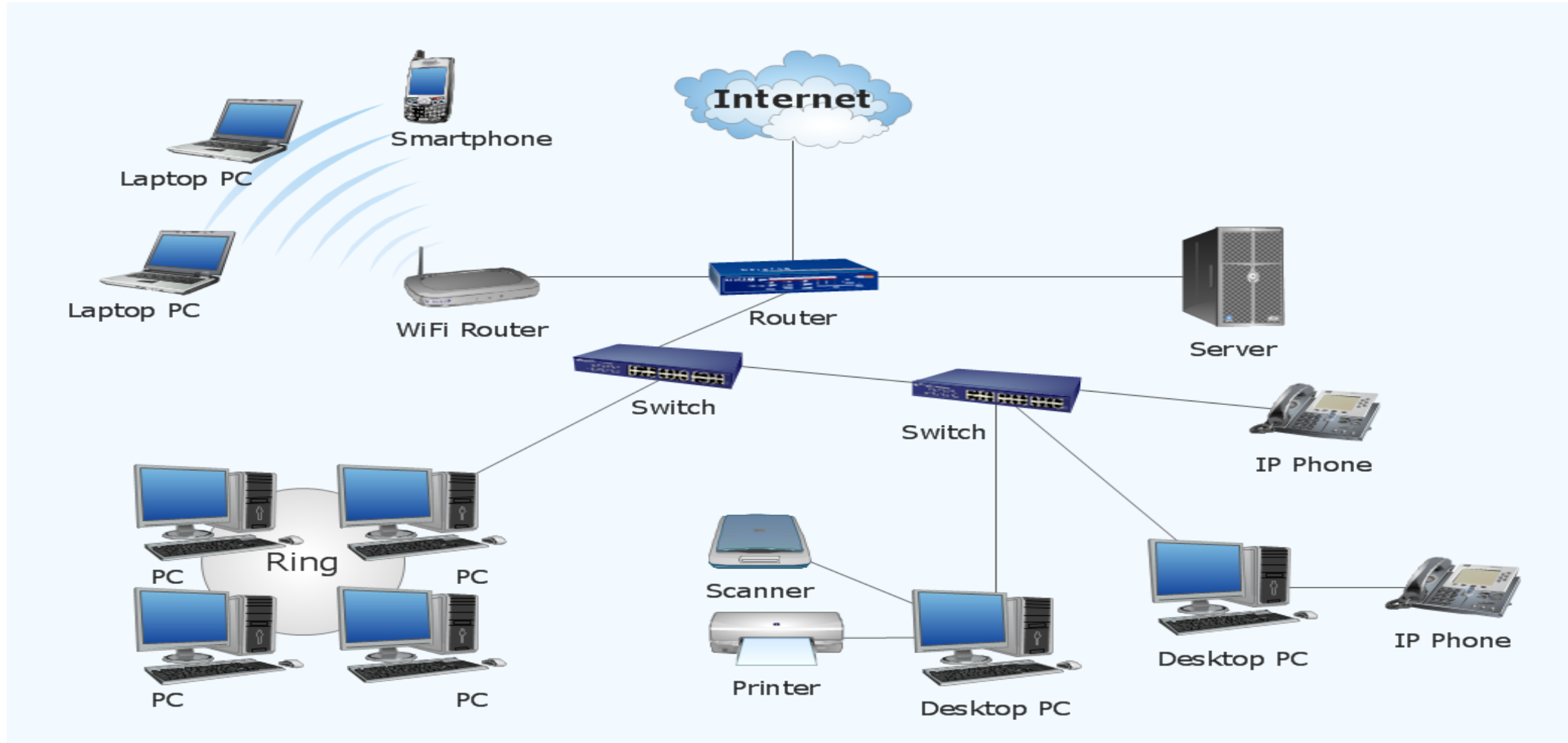
Building the network



What do you need to build your network?



Simple Network



Basic Network Elements (Hardware / Software)

– Hardware

– Devices

- Computers – Printers –Phone – Routers - Switches

– Medium

- Wired -Wireless –Satellites

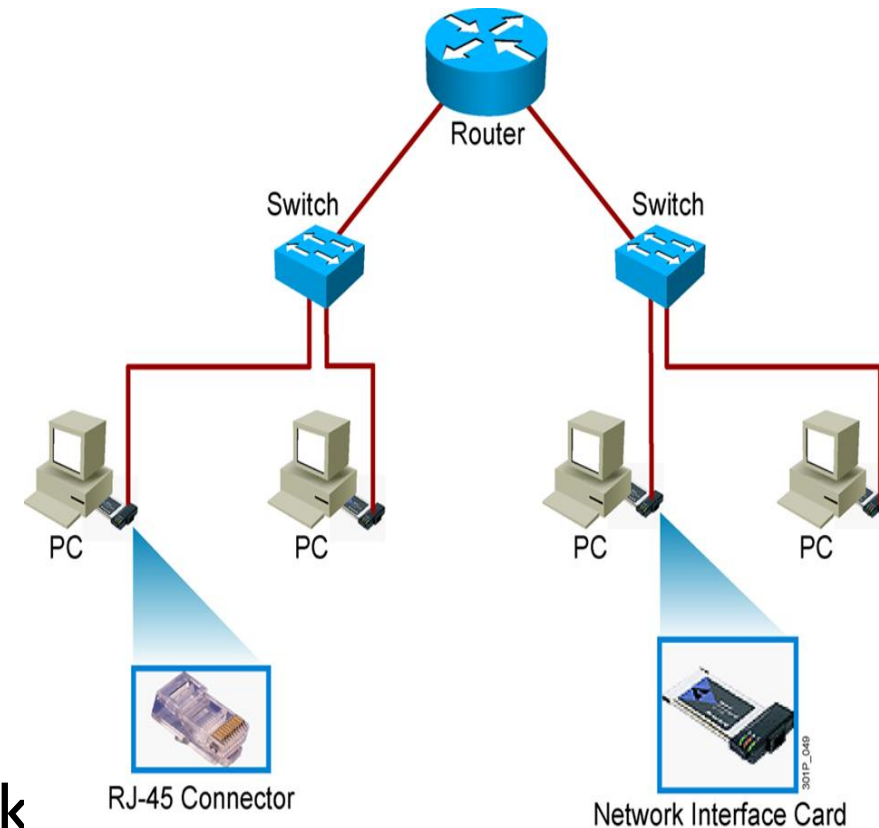
– Software

– Messages

- Information that travels over the medium
- Mails-WhatsApp....etc

– Protocols

- Governs how messages flow across network
- http –https-FTP-RDP



Basic Network Elements (Software)



Software
Protocols



Session 1 (TCP/IP Protocol Suite)

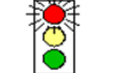
• Why we need Protocols ?

- To communicate efficiently
- Enable data to flow from one NIC to another
- Control the messages and the messages quantity in the network.

• OSI Reference Model

- **OSI:** Open Systems Interconnect
 - was defined by ISO in 1983
 - Give developers universal concepts so they can develop protocols
 - The OSI reference model breaks this approach into layers.



OSI MODEL			TCP / IP
7		Application Layer Type of communication: E-mail, file transfer, client/server.	FTP, SMTP, DNS, Telnet
6		Presentation Layer Encryption, data conversion: ASCII to EBCDIC, BCD to binary, etc.	
5		Session Layer Starts, stops session. Maintains order.	
4		Transport Layer Ensures delivery of entire file or message.	TCP (delivery ensured) UDP (delivery NOT ensured)
3		Network Layer Routes data to different LANs and WANs based on network address.	IP (ICMP, ARP, RARP)
2		Data Link (MAC) Layer Transmits packets from node to node based on station address.	
1		Physical Layer Electrical signals and cabling.	

Part 1_Basic Network Elements



Why a Layered Network Model?

- Reduces complexity
 - Easier to troubleshooting
- Standardizes interfaces
 - Multiple-vendor
- Layer Separation
 - Changes in one layer do not affect other layers
- Simplifies teaching and learning

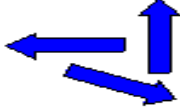
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1	Physical	 Physical Layer Electrical signals and cabling.	

Part 1_Basic Network Elements

OSI (7-Seven Layers)

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical



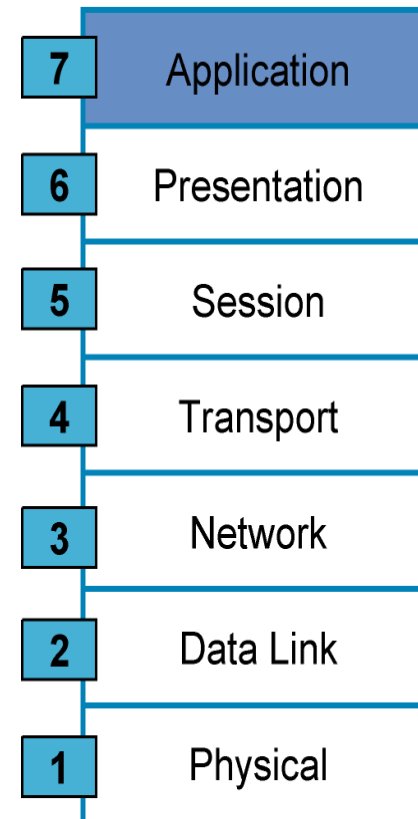
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Part 1_Basic Network Elements



◇ Application Layer

- ◇ Interface to end users
- ◇ provides Many services
 - ◇ File transfer
 - ◇ Network management
 - ◇ Email
- ◇ Protocols
 - ◇ HTTP (Hyper Text Transfer Protocol)
 - ◇ FTP (File transfer Protocol)
 - ◇ SMTP (Simple Mail transfer Protocol)
 - ◇ POP3 (Post office transfers Protocol)
 - ◇ Telnet/SSH (secure Shell)
- ◇ Example : Web browser



Network Processes to Applications

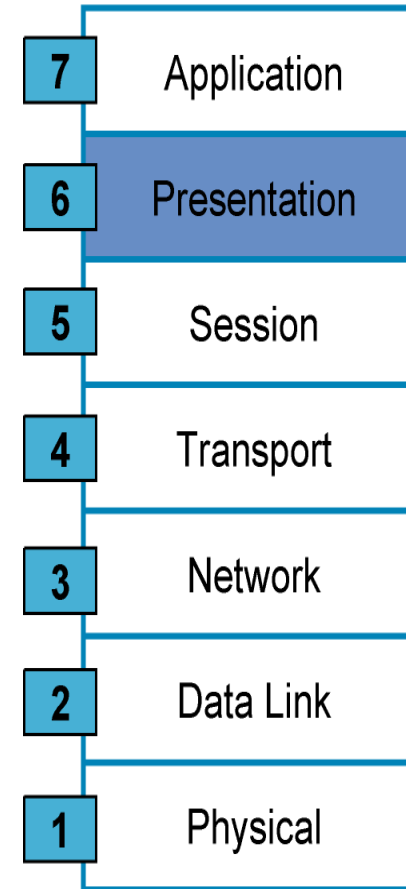
- Provides network services to application processes (such as electronic mail, file transfer, and terminal emulation)
- Provides user authentication

Part 1_Basic Network Elements

The Seven Layers Functions (Cont.)

◇ Presentation Layer

- ◇ Finding common presentation between source and Destination
- ◇ Ensures that is readable by receiving system
- ◇ **(support standardized application interface)**
- ◇ Formats data
- ◇ Structured data
- ◇ Negotiates data transfer syntax for application layer (Encoding and Decoding)
- ◇ Provides Encryption



Network Process to Applications

Data Representation

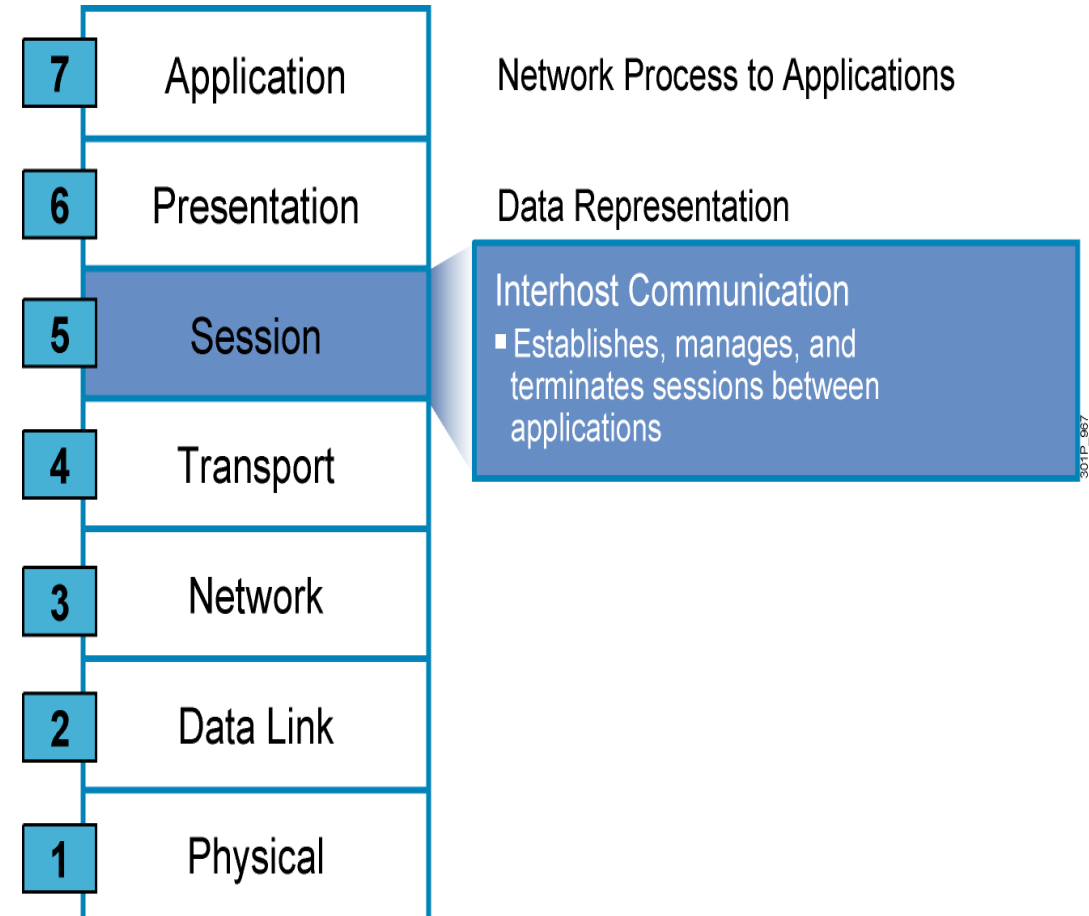
- Ensures that data is readable by receiving system
- Formats data
- Structures data
- Negotiates data transfer syntax for application layer
- Provides encryption

Basic Network Elements

The Seven Layers Functions (Cont.)

◇ Session Layer

- ◇ **Establishes, manages and terminates** sessions (connections) between cooperating applications (Dialogues)
- ◇ **Synchronization** (add checkpoints into a stream of data)
- ◇ **Controls the sessions** between the local and remote applications



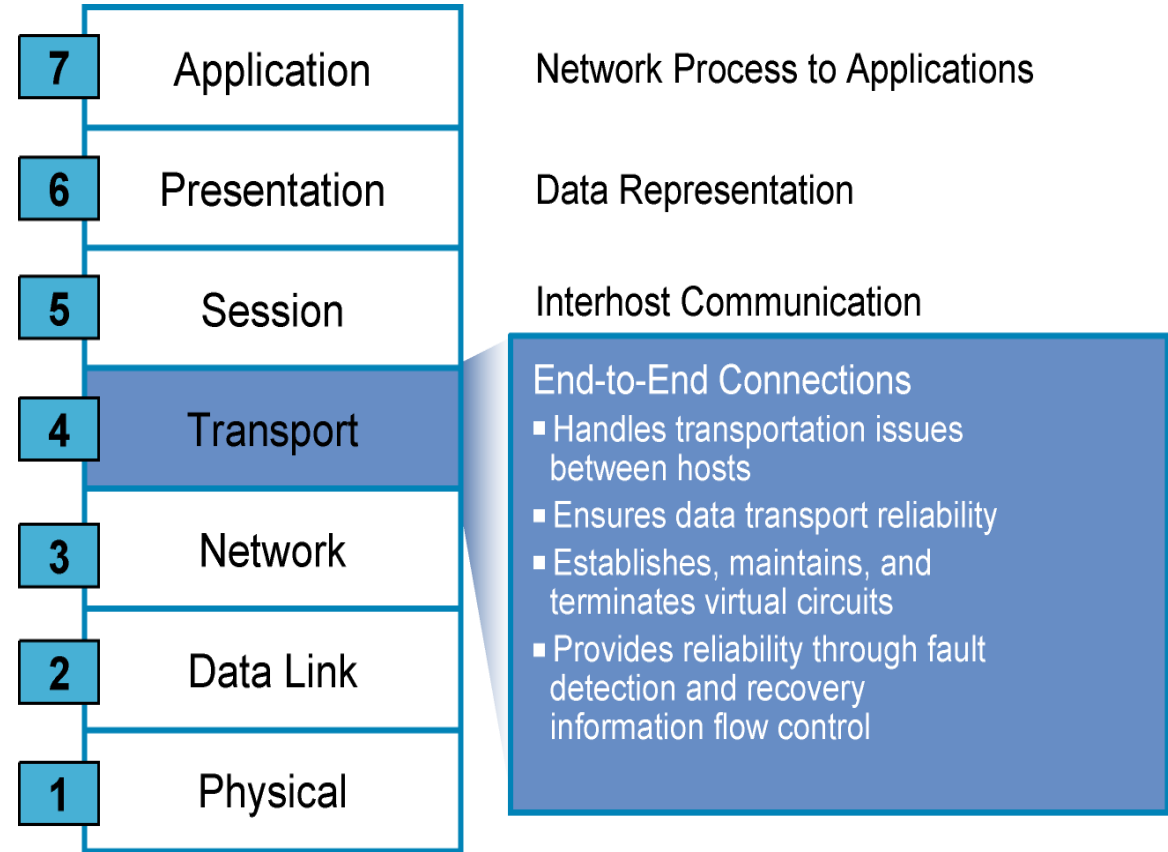
Part 1_Basic Network Elements



The Seven Layers Functions (Cont.)

◇ Transport Layer

- ◇ Organize data into **Segments**
- ◇ Provide **reliable** transport between end systems (source and destination hosts)
- ◇ End-to-end **error recovery**
- ◇ End-to-end **flow control**

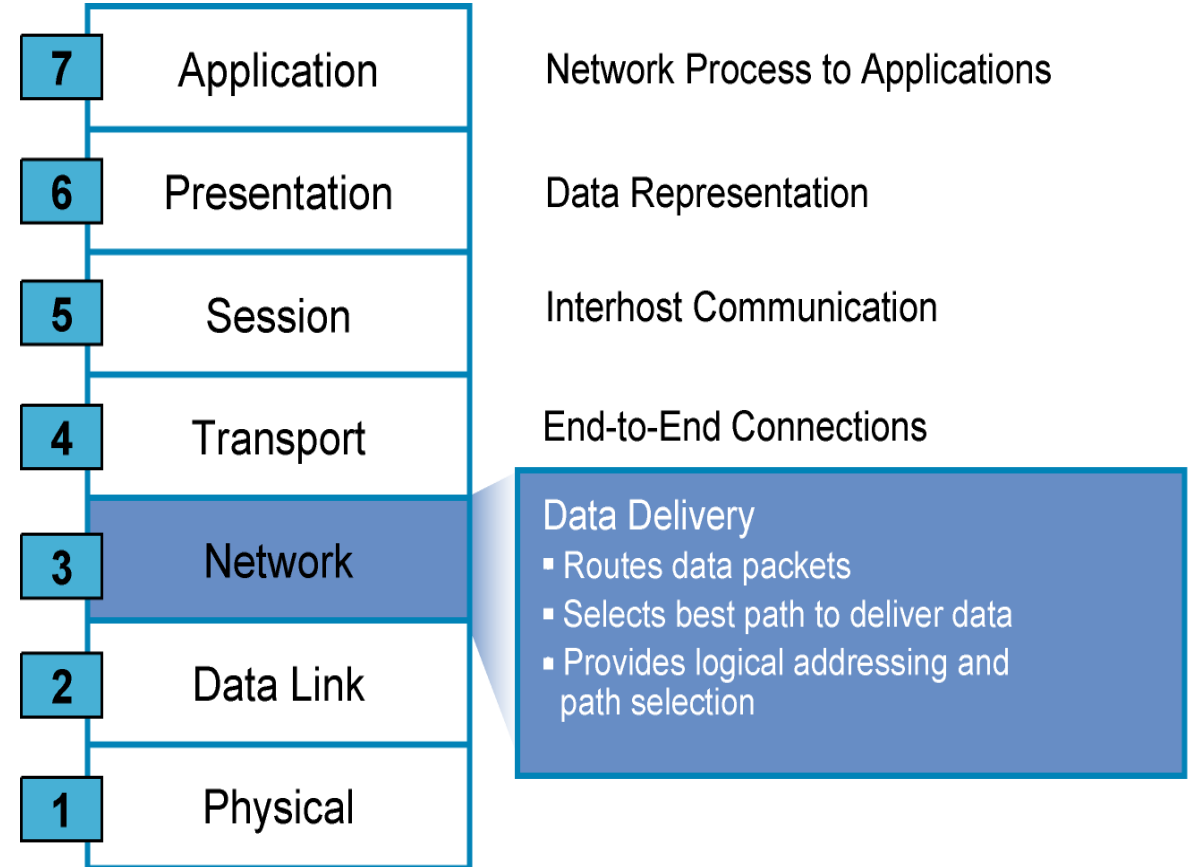


Part 1_Basic Network Elements

The Seven Layers Functions (Cont.)

◇ Network

- ◇ Organize data into **datagram** (**packets**)
- ◇ It is responsible for the Internet Protocol **Addressing (IP)** (Addressing)
- ◇ It know the **best path** for the destination (**Routing**)
- ◇ End-to-end Addressing

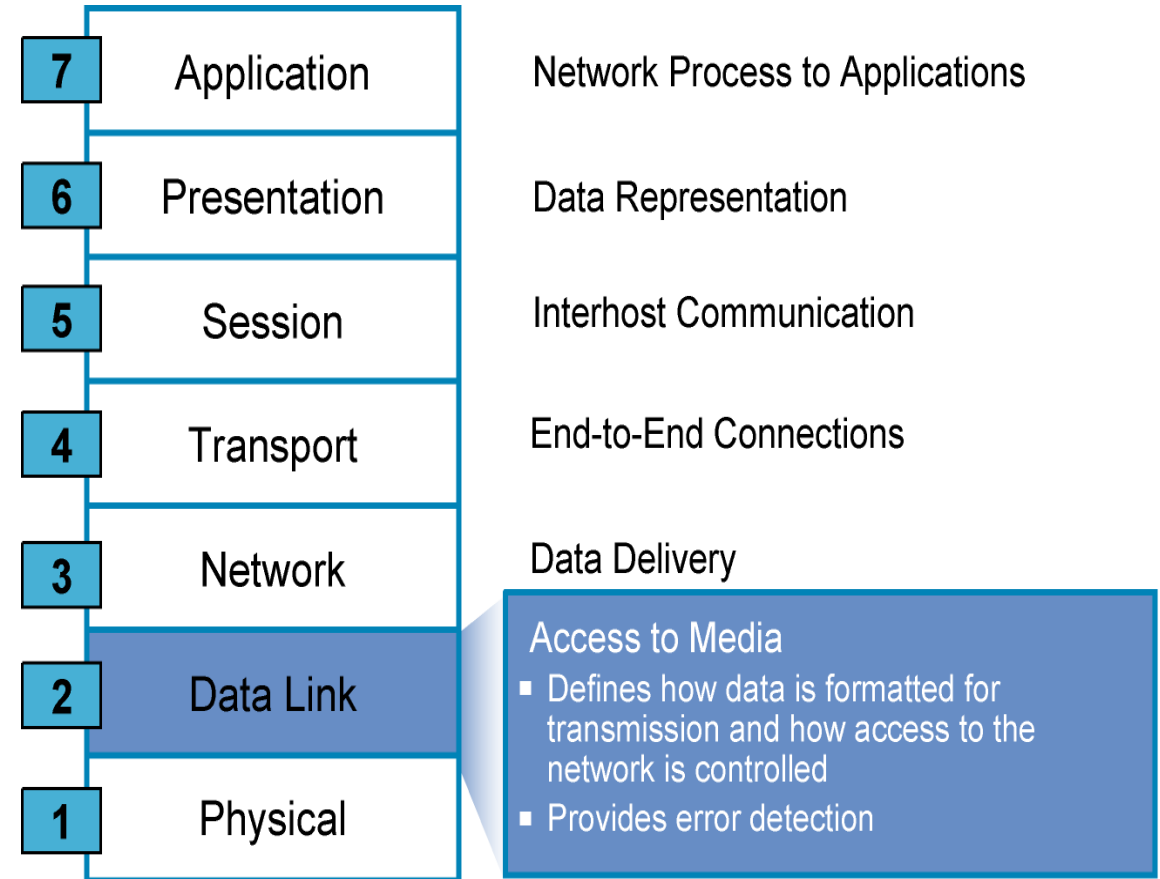


Part 1_Basic Network Elements

The Seven Layers Functions (Cont.)

◇ Data link

- ◇ Reliable data transfer across a **physical link** (Error Control)
- ◇ Organize the data into **Frames**, to be put on the physical medium
- ◇ Check the Frame For errors
- ◇ **Hop to hop** addressing

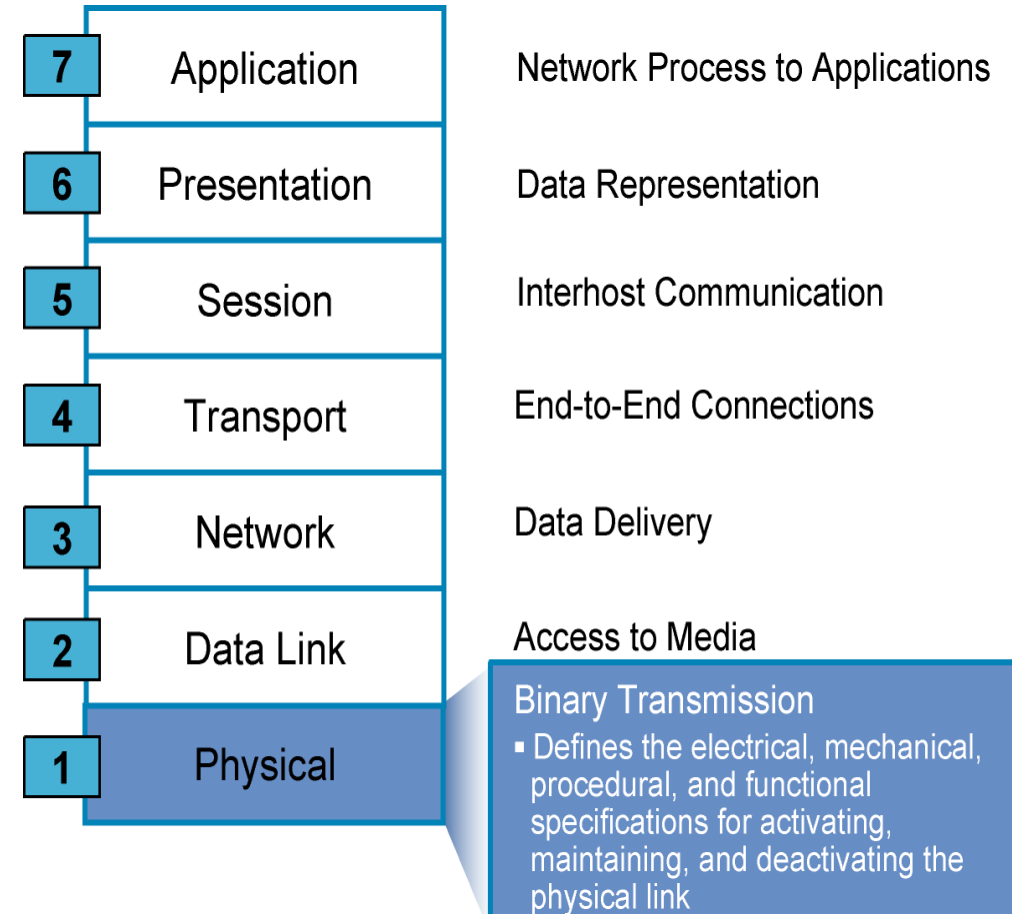


Part 1_Basic Network Elements

The Seven Layers Functions (Cont.)

◇ Physical

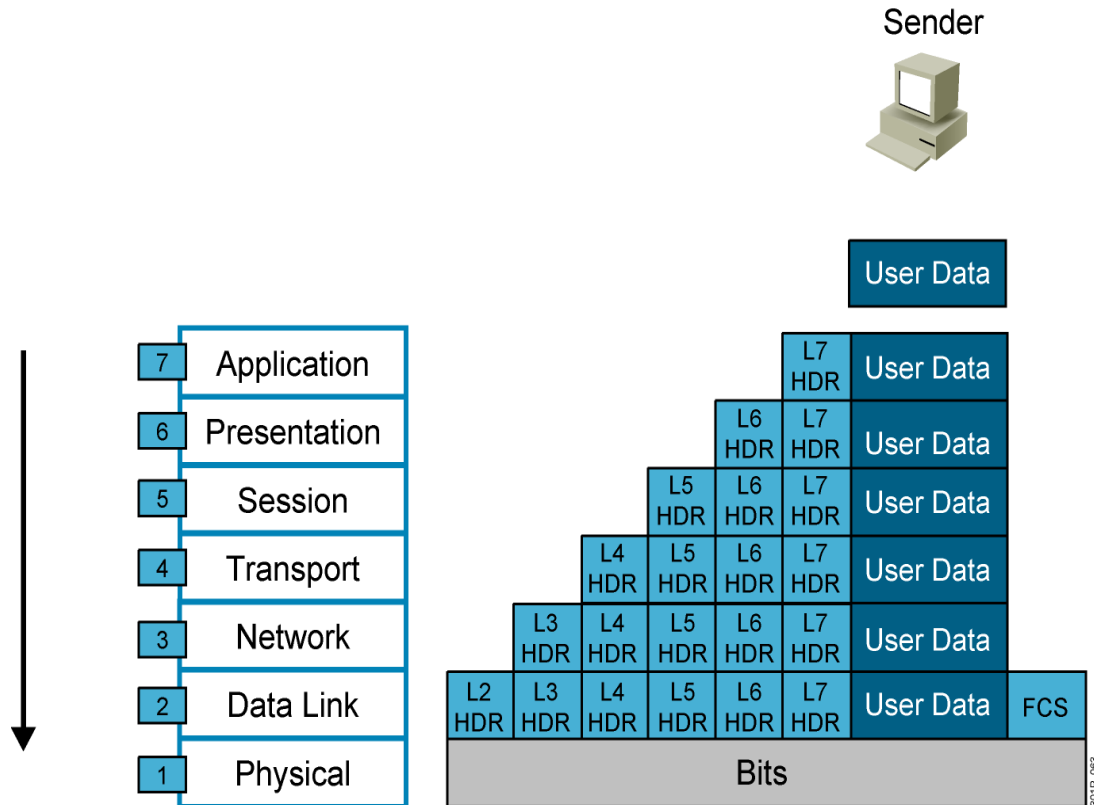
- ◇ Transmission of unstructured **bit stream** over the physical link
- ◇ Deals with the mechanical and electrical specifications of the interface and transmission media (cables and connectors)
- ◇ Representation of bits



Part 1_Basic Network Elements

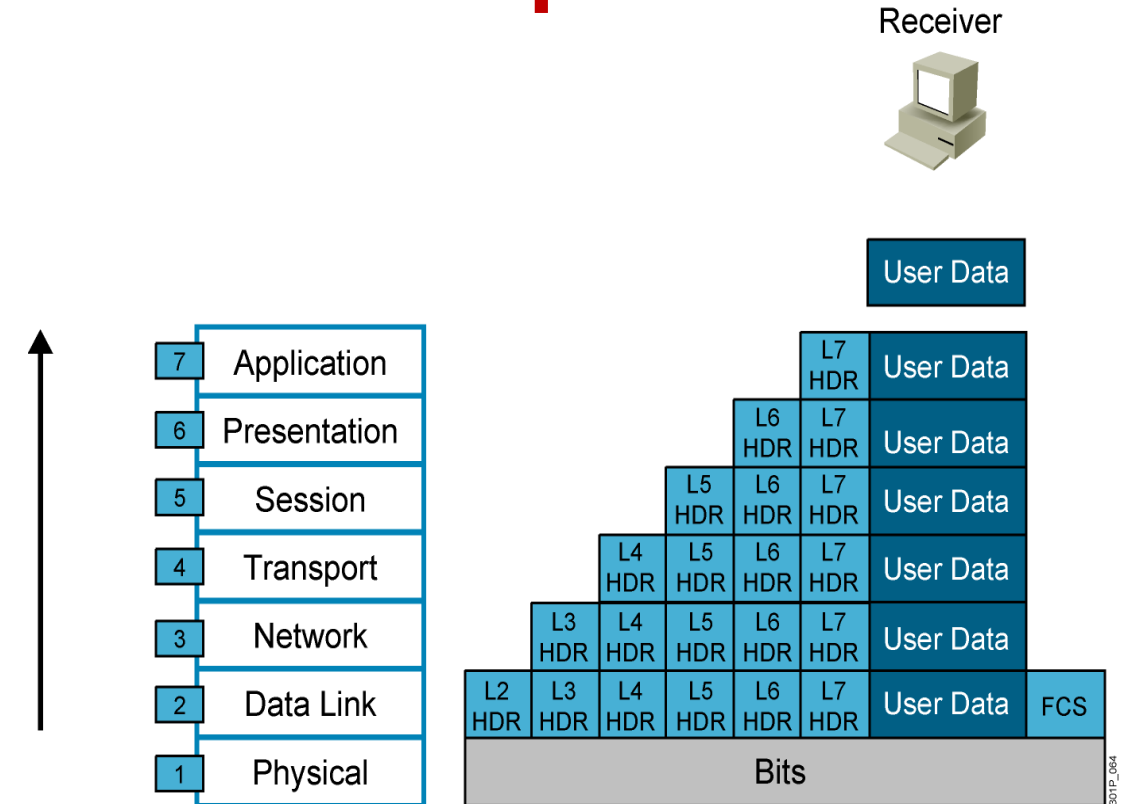


Data Encapsulation



HDR = Header

Data De-Encapsulation



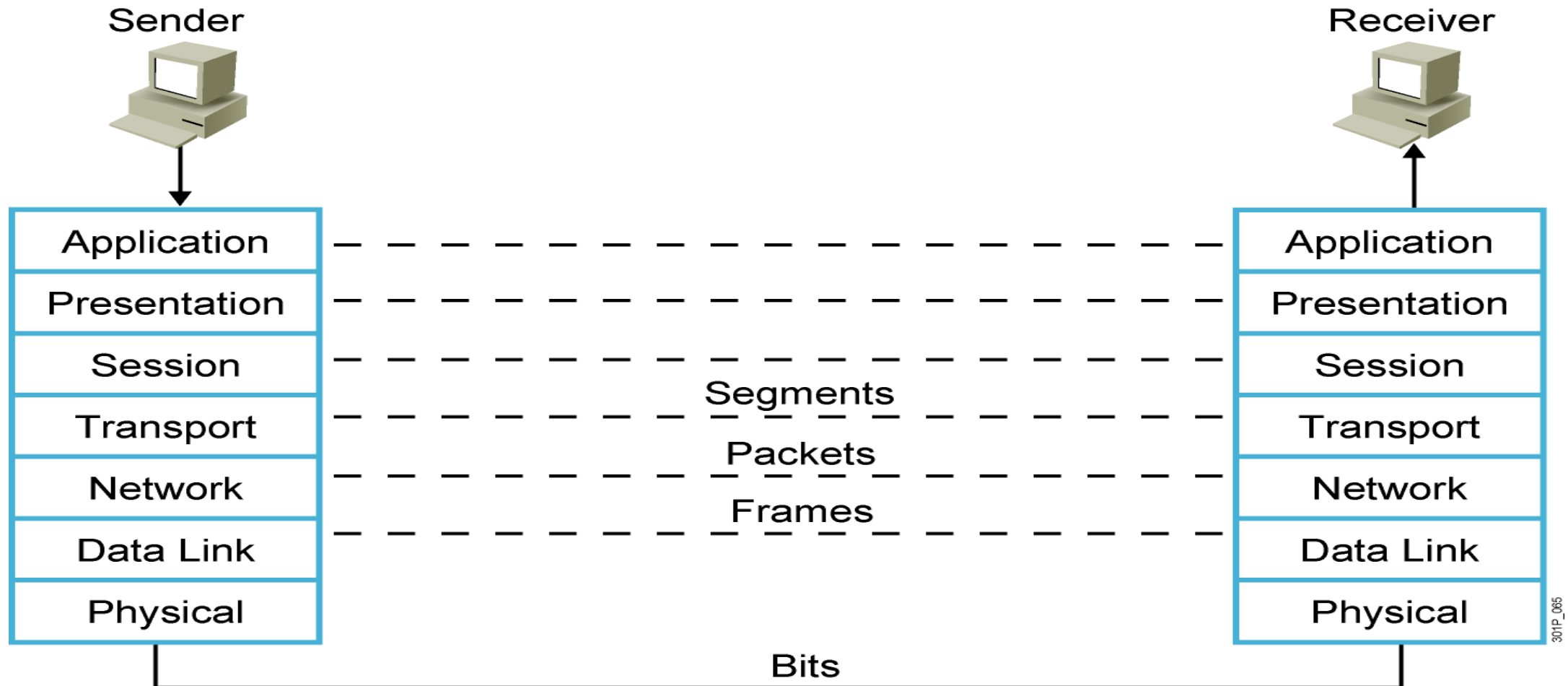
HDR = Header

(FCS) frame check sequence used for error-detecting/Hash

Part 1_Basic Network Elements



Peer-to-Peer Communication



Part 1_Basic Network Elements



Transmission Control Protocol/Internet Protocol TCP /IP



Session 1 (TCP/IP Protocol Suite)



- **TCP/IP**

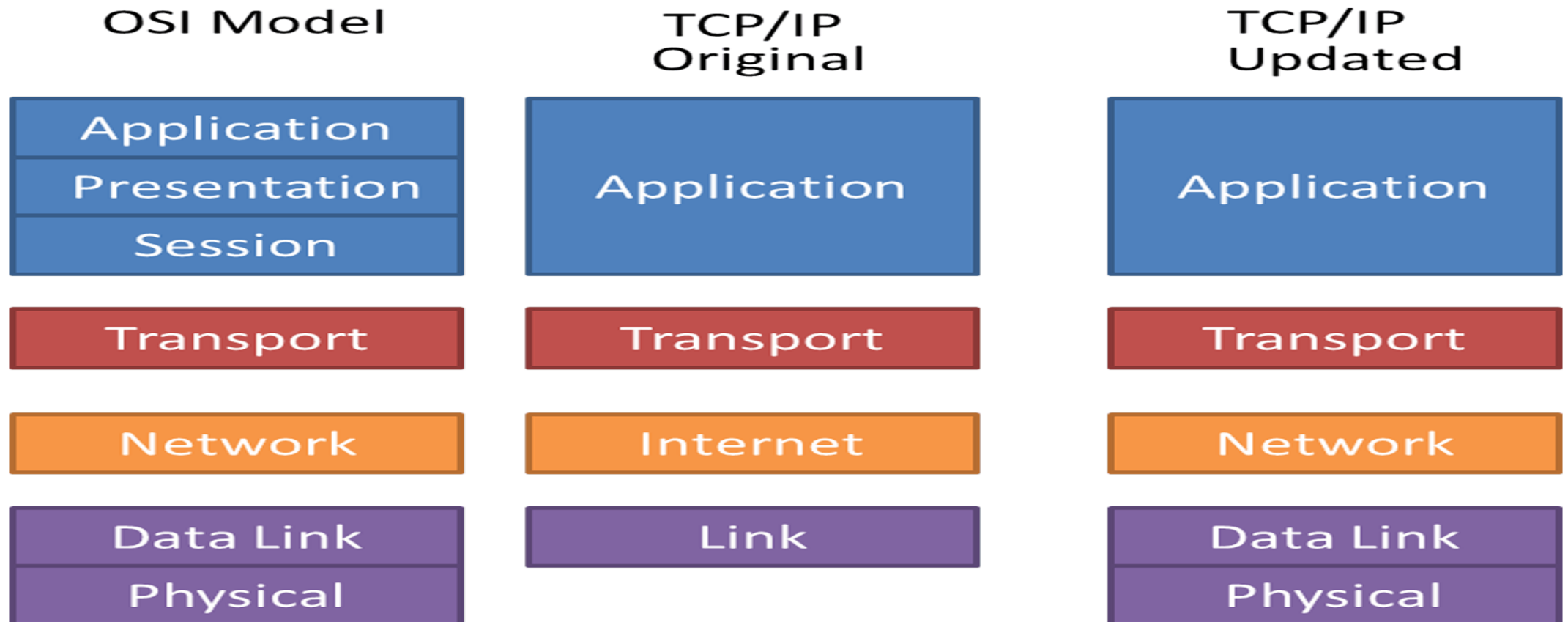
- **Transmission Control Protocol/Internet Protocol.**
 - **Open standard protocol**
 - **Cross Platform** (default protocol for all modern operating systems)
 - Microsoft Operating Systems
 - LINUX Operating Systems
 - **Not tied to one vendor**
 - **Direct access to the Internet**(TCP/IP is the internet protocol)
 - Now internet use TCP/IP v4
 - Next version TCP/IP v6
 - **Routable**



Session 1 (TCP/IP Protocol Suite)

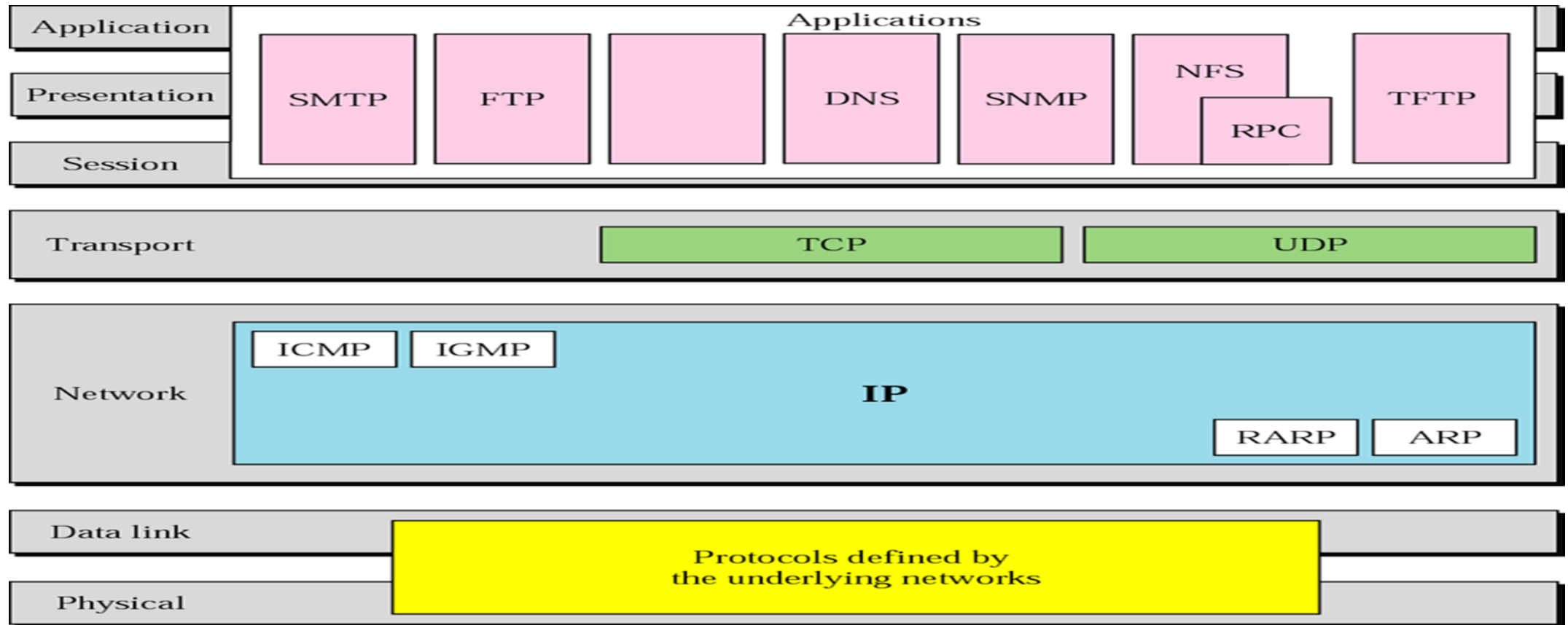


- **TCP/IP VS. OSI Model**



(TCP/IP Protocol Architecture)

- Some Protocols in TCP/IP Suite



Session 1 (TCP/IP Protocol Architecture)



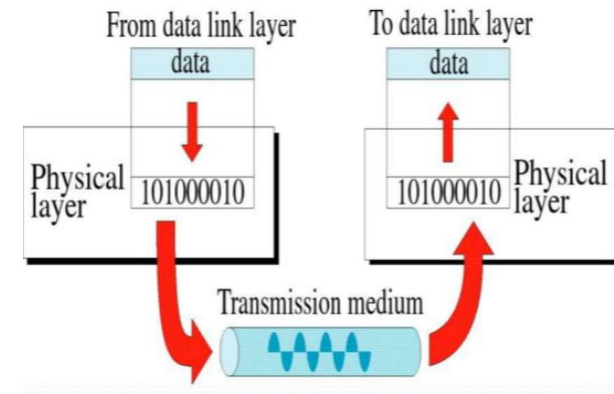
- **Network Access Layer**

- **Physical Layer**

- defines the electrical, Transmission medium
 - movements of individual **Bits** from one node to next

- **Datalink Layer**

- Logical interface between end system and network
 - Error notification.
(FRAMES, MAC(Media Access Control))
 - Hop to Hop addressing
 - Error detection Mechanism (detects damaged or lost frames)

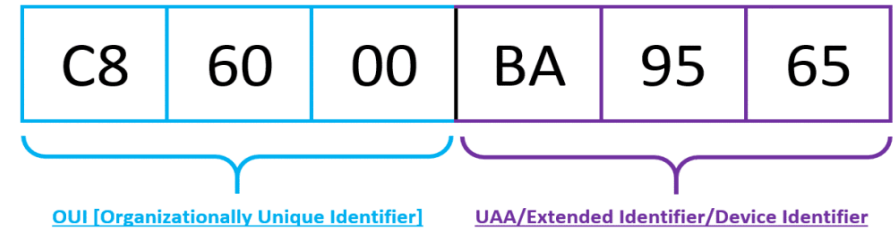


Session 1 (TCP/IP Protocol Architecture)

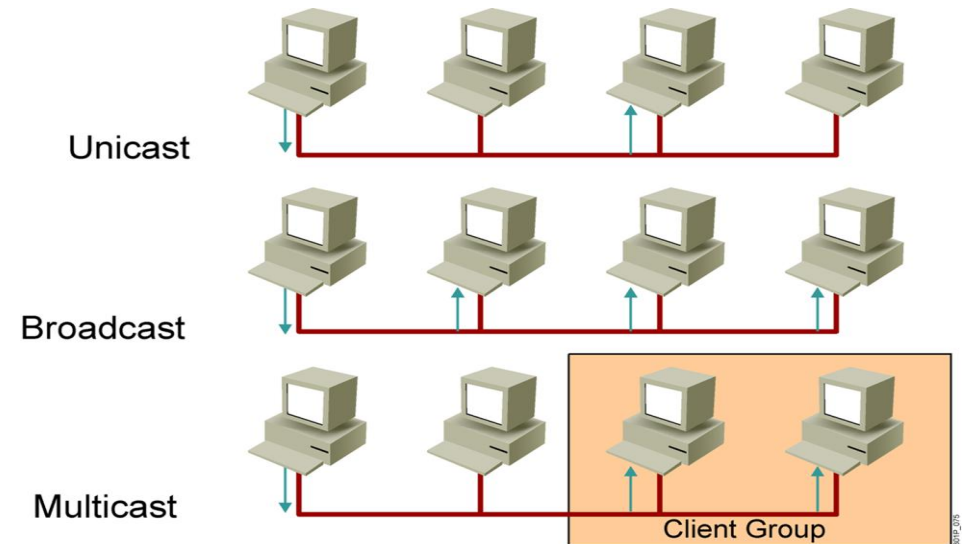


- **Physical Addresses (Mac)**

- Physical Address burned on the card
- Unique address over the world
- 48-bit (6-byte) written as 12 hexadecimal digits;
- every byte (2 hexadecimal digits) is separated by a colon



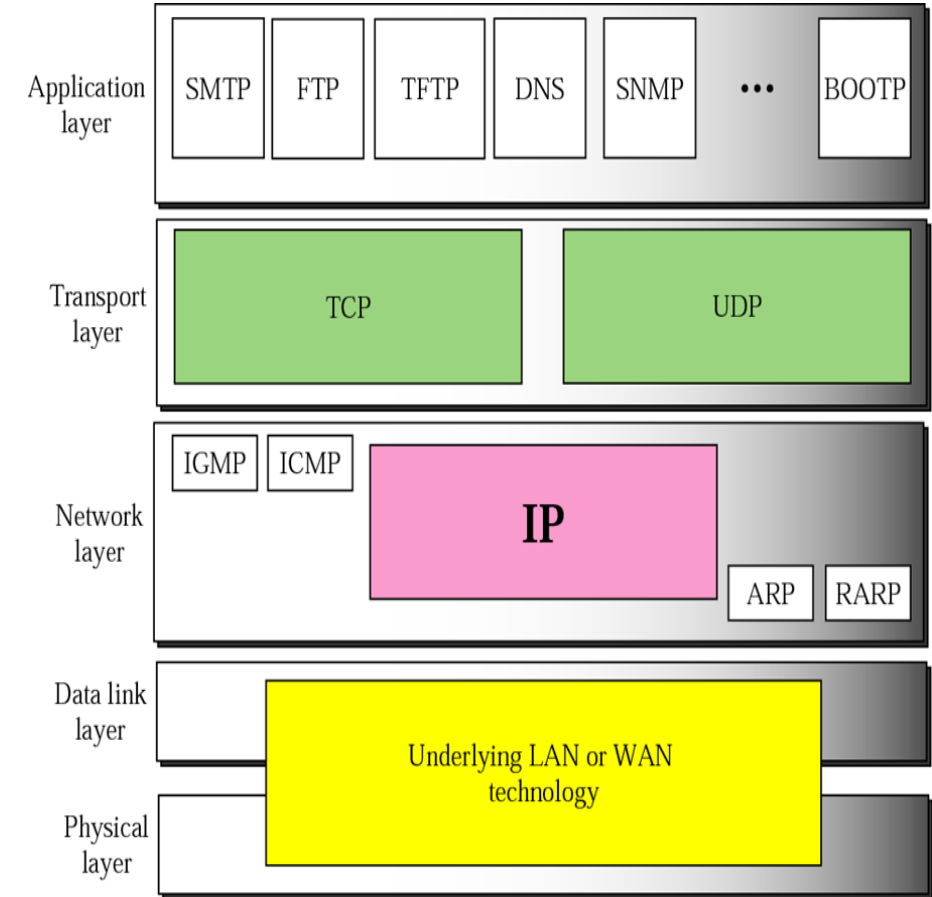
- Physical addresses can be either
 - **Unicast**
 - **Multicast**
 - **Broadcast**
- To check your physical address: -
 - **Ipconfig /all**
 - **GetMac**



Session 1 (TCP/IP Protocol)

- **Internet / Network protocol Layer (IP Layer)**

- Provides connectivity and path selection between two hosts (Source to Destination)
- Routing of data (Provide mechanism to transmit data over independent networks that are linked together)
- Logical addressing IPV4 , IPV6



Session 1 (TCP/IP Protocol Architecture)



- **Internet Protocol (IP V4)**

- Uniquely identify each device on an IP network layer.
- Some times we called it the logical address
- Every host (computer, networking device, peripheral) must have a unique address at the same network
- The IP address 32 bit divided into 4 octets each octet 8 bit

1 octet = 8 bit each represents from 0 to 255 separated with dots

An IP address is a 32-bit binary number	Example			
	10101100	00010000	10000000	00010001
For readability, the 32-bit binary number can be divided into four 8-bit octets	10101100	00010000	10000000	00010001
Each octet (or byte) can be converted to decimal	172	16	128	17
The address can be written in dotted decimal notation	172.	16.	128.	17

•The address space of IPv4 is 2³² or 4,294,967,296



Part 1 (TCP/IP Protocol Architecture)



IP ADDRESS RANGES The First Octet

Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0
D	224.0.0.0 to 239.255.255.255	Reserved for Multicasting
E	240.0.0.0 to 254.255.255.255	Experimental

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback testing.

A B C ... Easy as 1 2 3

Class **A** ... First 1 bit fixed 0 x x x x x x x . Host . Host . Host

Class **B** ... First 2 bits fixed 1 0 x x x x x x . Network . Host . Host

Class **C** ... First 3 bits fixed 1 1 0 x x x x x . Network . Network . Host

Session 1 (TCP/IP Protocol Architecture)



PUBLIC IP ADDRESSES (Real IP) Private IP Addresses (Local IP)

Class	Public IP Ranges
A	1.0.0.0 to 9.255.255.255 11.0.0.0 to 126.255.255.255
B	128.0.0.0 to 172.15.255.255 172.32.0.0 to 191.255.255.255
C	192.0.0.0 to 192.167.255.255 192.169.0.0 to 223.255.255.255

Class	Private Address Range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255

- **Nat** is used to Translate the private IP address to public IP addresses.



Session 1 (TCP/IP Protocol Architecture)



- **How to assign IP address to device**

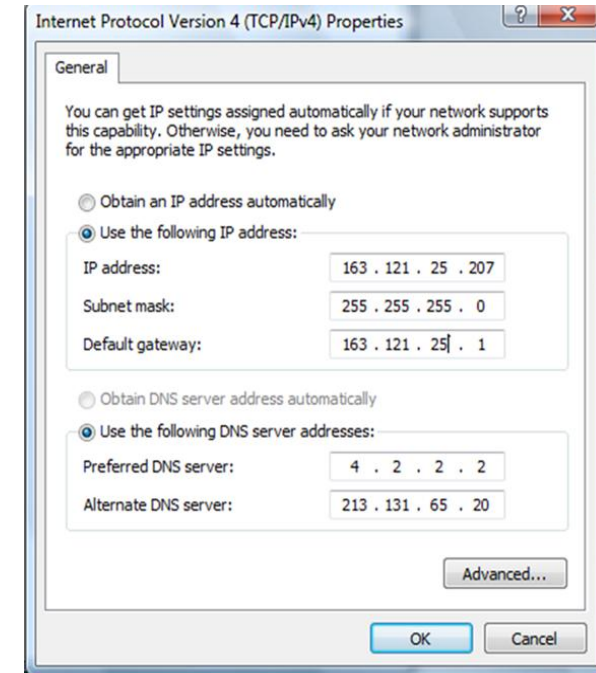
- Manually
- Automatic (By DHCP)
- APIPA (Random /Rang : 169.254.X.X)

- **To find your private IP**

- `Ipconfig - Ipconfig /all - Ipconfig /release - Ipconfig /renew`

- **ICMP (Ping) To test connectivity between Hosts**

- Ping IP
- Ping URL
- Ping IP `-l -n -t`

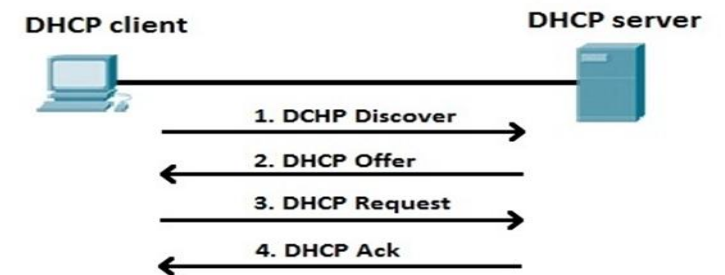
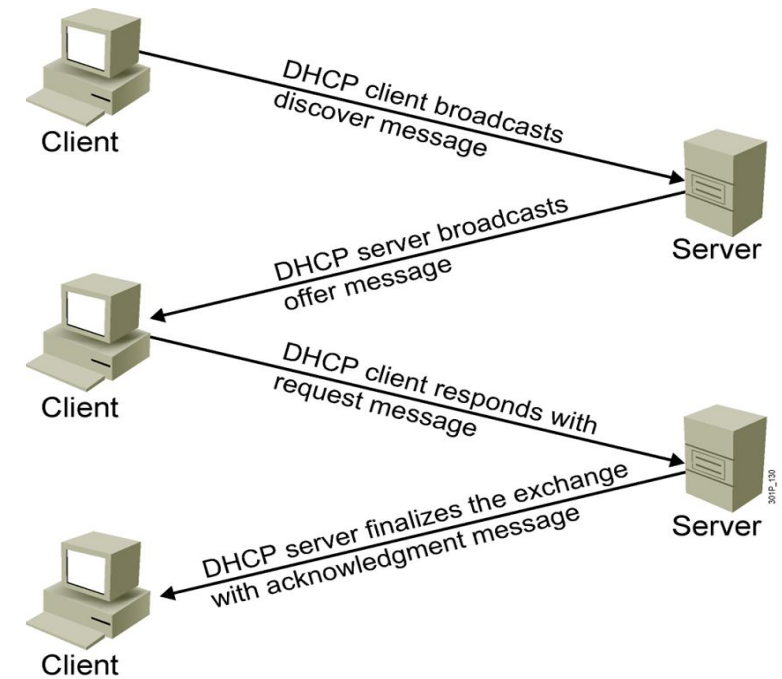


Part 1 (TCP/IP Protocol Architecture)-

❖ Assign IP address Automatic

■ DHCP Server

- used to assign **dynamically** the IP Configuration including
 - Host IP
 - Subnet mask
 - Default Gateway
 - DNS server IP
 - Lease Time
- **Dora** (Discover –Offer-Request-Acknowledgment)



Part 1 (TCP/IP Protocol Architecture)-



❖ Assign IP address Automatic

■ APIPA

■ APIPA

- If **no DHCP server** is available the APIPA is used
- Auto configuration IP address
- let LAN users talk to Each other if the DHCP fails
- Can **not** be Routed
- **Rang : 169.254.X.X**

```
Link-local IPv6 Address . . . . . : fe80::3428:a83b:6e6b:24b2%3
Autoconfiguration IPv4 Address. . : 169.254.36.178
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
```



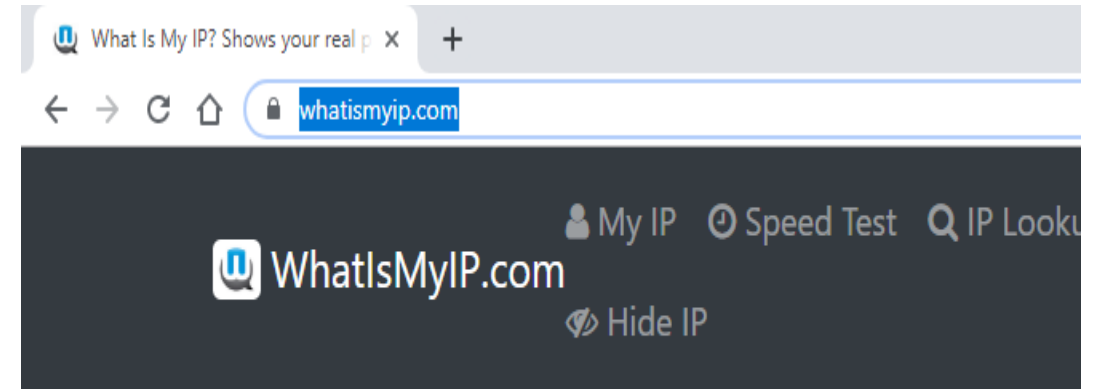
Part 1 (TCP/IP Protocol Architecture)-

- To know your real IP addresses

<https://www.whatismyip.com/>

- To get bulk of the Public IP address you get it from your Internet service provider

- With the grow the of the pubic IP address we used **NAT** and **IPV6**



Your Public IPv4 is: 41.236.145.96

Your IPv6 is: Not Detected

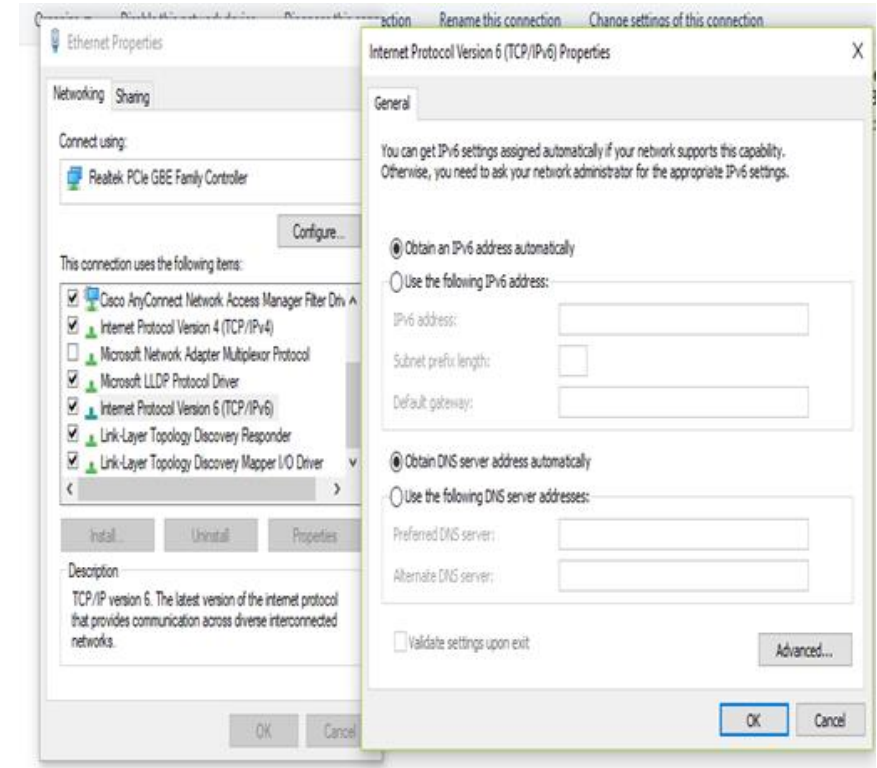
Location: Al Jizah, GZ EG ?

ISP: TE Data

Session 1 (TCP/IP Protocol Architecture)

- **Internet Protocol (IP V6)**

- **128-bit** address, provides approximately (340,282,366,920,938,463,374,607,431,768,211,456) = 340 billion billion billion billion, addresses)
- Represented as eight groups, separated by colons, of four hexadecimal digits. The full representation may be simplified by several methods of notation;



2001:0db8:0000:0000:0000:8a2e:0370:7334

=

2001:db8::8a2e:370:7334

Session 1 (TCP/IP Protocol Architecture)



- **Internet Of Things (IOT)**

- Aims connect all devices to the existing Internet infrastructure.
- "things" that sense and collect data and send it to the internet.
- (Eg:- coffee maker, A.C, Washing Machine, Ceiling Fan, lights , any thing) having sensors can be connected with internet.

- **PRACTICAL APPLICATIONS:-**

- Smart Homes -Smart Cities-Energy - Environment monitoring- healthcare- Management



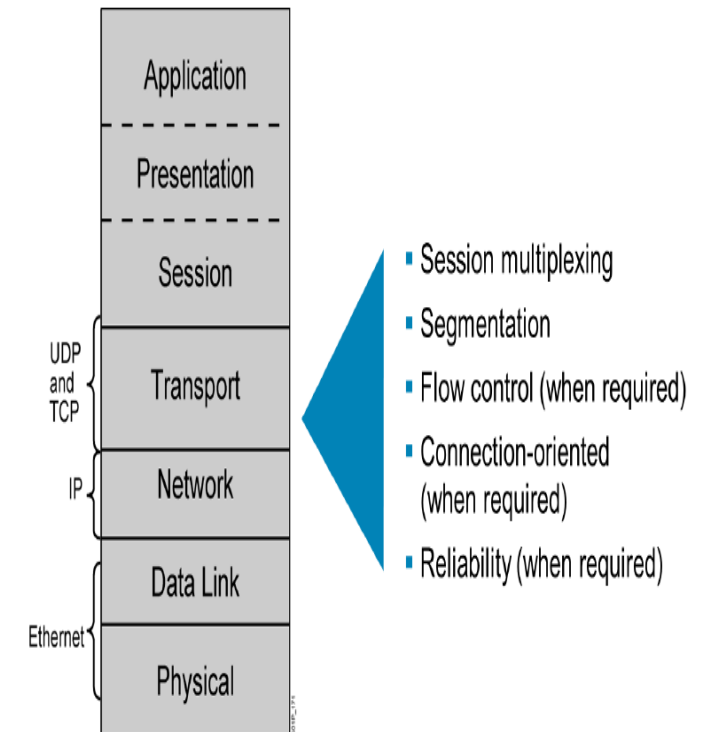
Basic Network Elements (Software)



Transport Layer

- **Session multiplexing:**
 - open **multiple sessions** using **UDP and TCP**

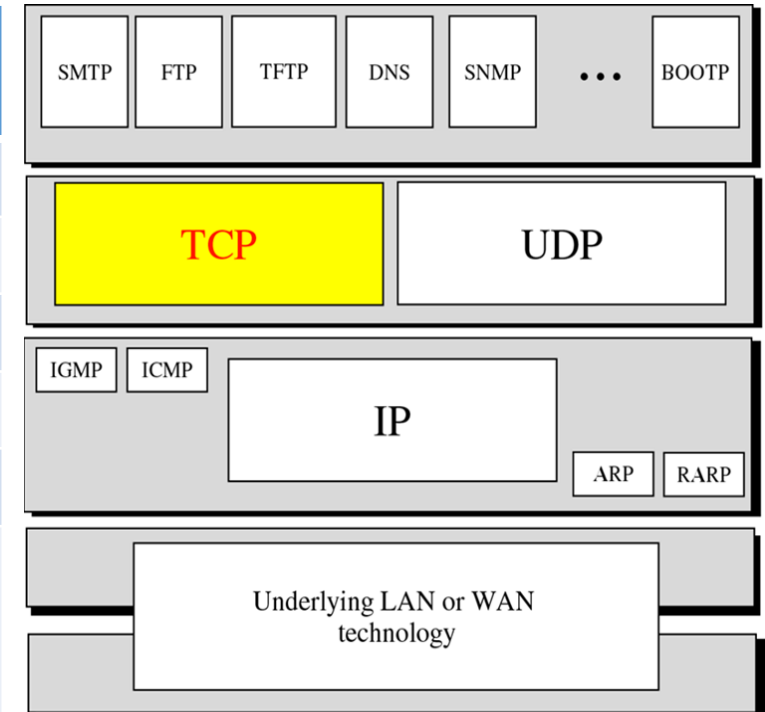
Example : open **cisco.com** & open **facebook.com** you are the source using **port 49001** and another **port 49002** on the same machine (session multiplexing).
- **Segmentation:**
 - divided the data up to **multiple segments** to be easier in handling (the maximum performance 1518 byte)
- **Connection Oriented:**
 - To maintain the session with **acknowledgements** that the data are sent to the receiver then terminate the session
- **Reliability:**
 - **Data corrections** and **avoid the duplicate data** out of order and data arrangement.



Session 1 (TCP/IP Protocol Architecture)

• Transport Layer

TCP Transmission Control Protocol)	UDP User Datagram Protocol
Reliable (Acknowledgement)	Unreliable (Best –Effort delivery)
Connection oriented (synchronization)	Connectionless (no notification)
Full duplex	Full duplex
Error control(Error checking(checksum)	Perform very limited error checking
Data-recovery features	Has no Data-recovery features
E-mail File sharing Downloading	Voice Streaming Video Streaming

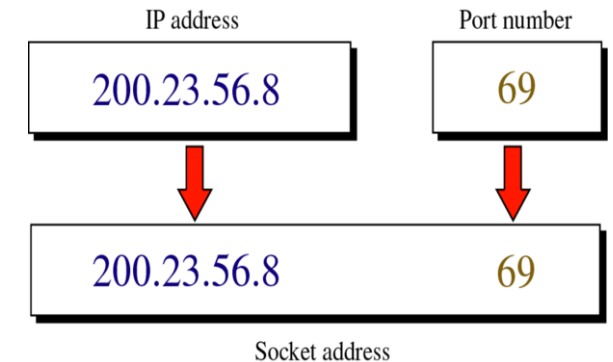
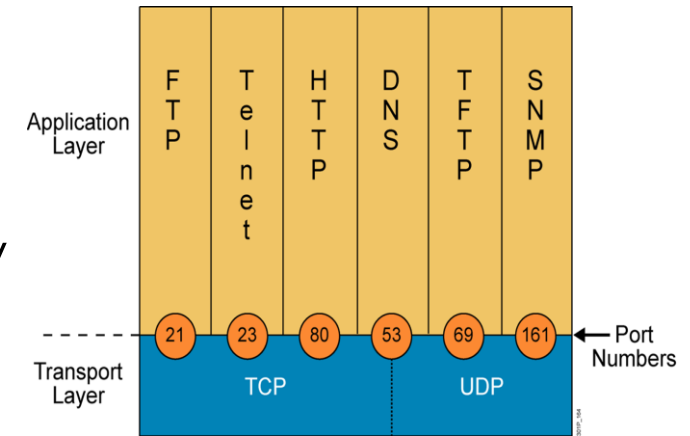


Session 1 (TCP/IP Protocol Architecture)



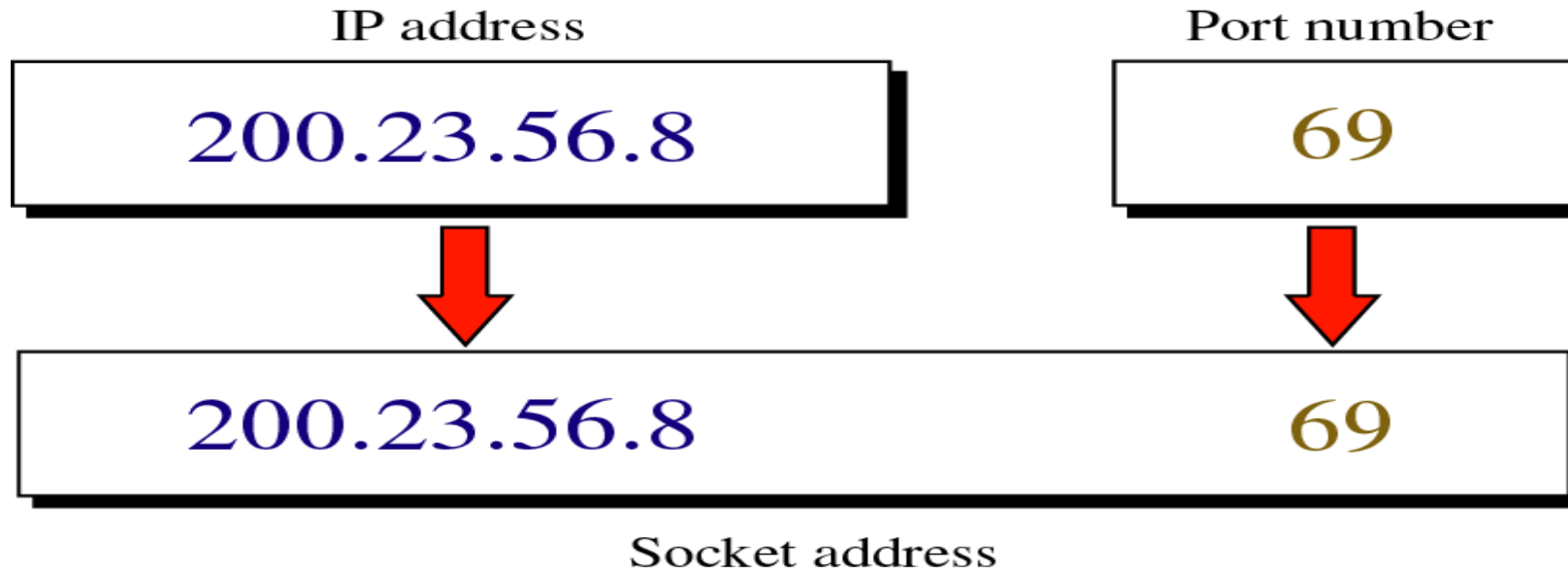
- **Transport Layer addressing (Port Numbers.)**

- (ICANN) controls the port numbers.
- **Well Known ports**
 - permanent used numbers.
Range from 0 to 1,023 are assigned and controlled by ICANN
- **Registered ports**
 - Range from 1,024 to 49,151 not assigned or controlled by ICANN
but can be registered at ICANN to avoid duplication
- **Dynamic ports**
 - Range from 49,152 to 65,535 are neither controlled nor registered



Part 1 (TCP/IP Protocol Architecture)

Socket Address



Part 1 (TCP/IP Protocol Architecture)-



❖ NETSTATE

netstat -n

netstat -a

To know session and ports on your device

```
Command Prompt
^C
C:\Users\ITD-mabdsalam>netstat -n

Active Connections

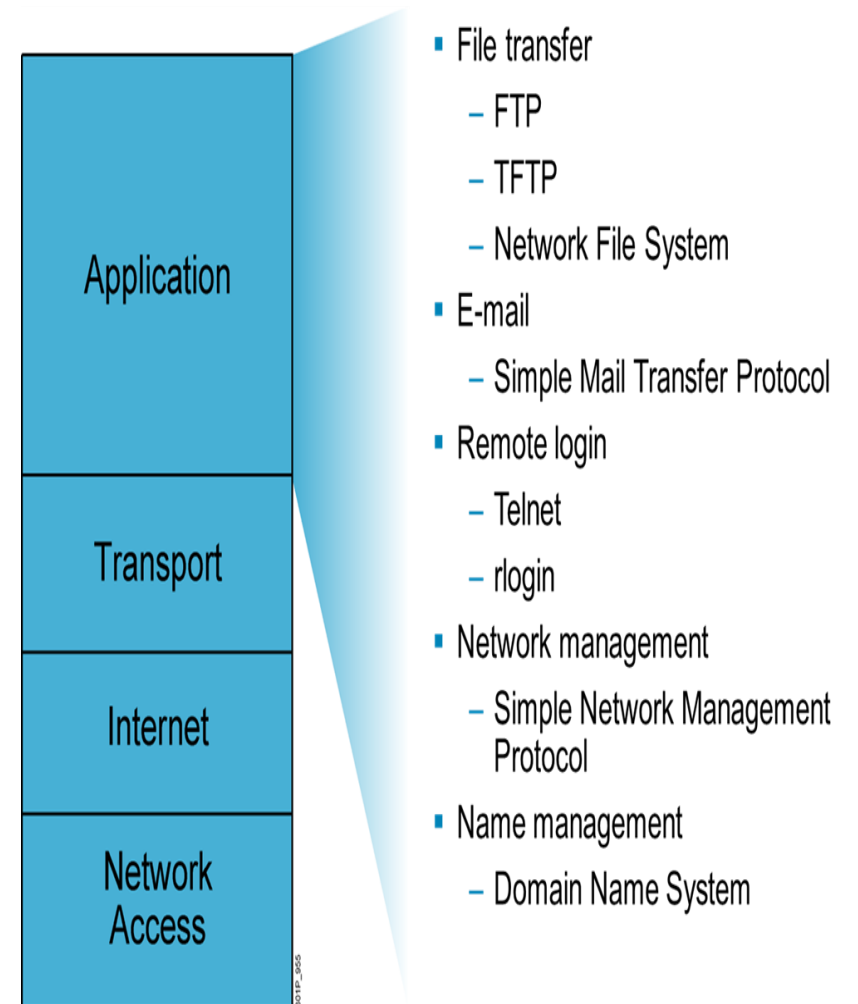
Proto Local Address          Foreign Address         State
TCP    127.0.0.1:40000          127.0.0.1:49673        ESTABLISHED
TCP    127.0.0.1:49673        127.0.0.1:40000        ESTABLISHED
TCP    127.0.0.1:52786        127.0.0.1:60014        ESTABLISHED
TCP    127.0.0.1:52787        127.0.0.1:60012        ESTABLISHED
TCP    127.0.0.1:60010        127.0.0.1:60158        ESTABLISHED
TCP    127.0.0.1:60012        127.0.0.1:52787        ESTABLISHED
TCP    127.0.0.1:60014        127.0.0.1:52786        ESTABLISHED
TCP    127.0.0.1:60014        127.0.0.1:60157        ESTABLISHED
TCP    127.0.0.1:60154        127.0.0.1:62522        ESTABLISHED
TCP    127.0.0.1:60156        127.0.0.1:62522        ESTABLISHED
TCP    127.0.0.1:60157        127.0.0.1:60014        ESTABLISHED
TCP    127.0.0.1:60158        127.0.0.1:60010        ESTABLISHED
TCP    127.0.0.1:62522        127.0.0.1:60154        ESTABLISHED
```

Session 1 (TCP/IP Protocol Architecture)



- **Application Layer**

- Communication between processes or applications
- **Internet Services (Client/Web Server)**
 - The World Wide Web: HTTP
 - Naming Service: DNS
 - File Transfer: FTP
 - Telnet Service
 - Electronic Mail service: IMAP, POP3, SMTP



Session 1 (TCP/IP Protocol Architecture)



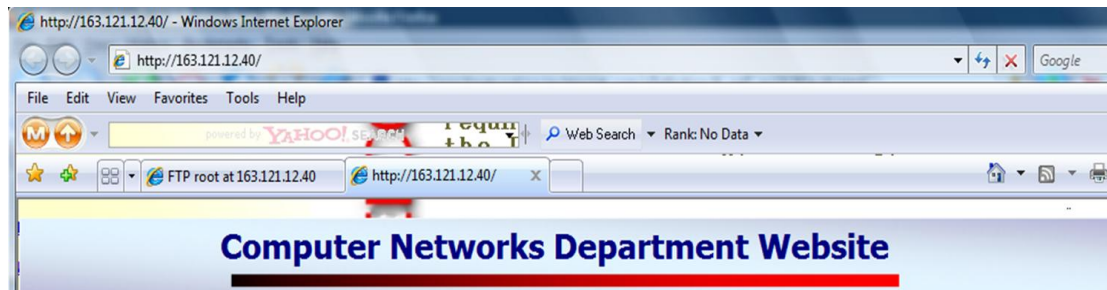
• URL is Universal Resource Locator



- **Protocol** : HTTP, HTTPS or FTP
- **Host** : is the domain name of the computer on which the information is located .
- **Port**: The URL can optionally contain the port number of the server
- **Path**: is the pathname of the file where the information is located.

• HTTP (Hyper Text Transfer Protocol)

- Supports the delivery of web pages to the client



Session 1 (TCP/IP Protocol Architecture)

- **DNS** (Domain Name Servers)

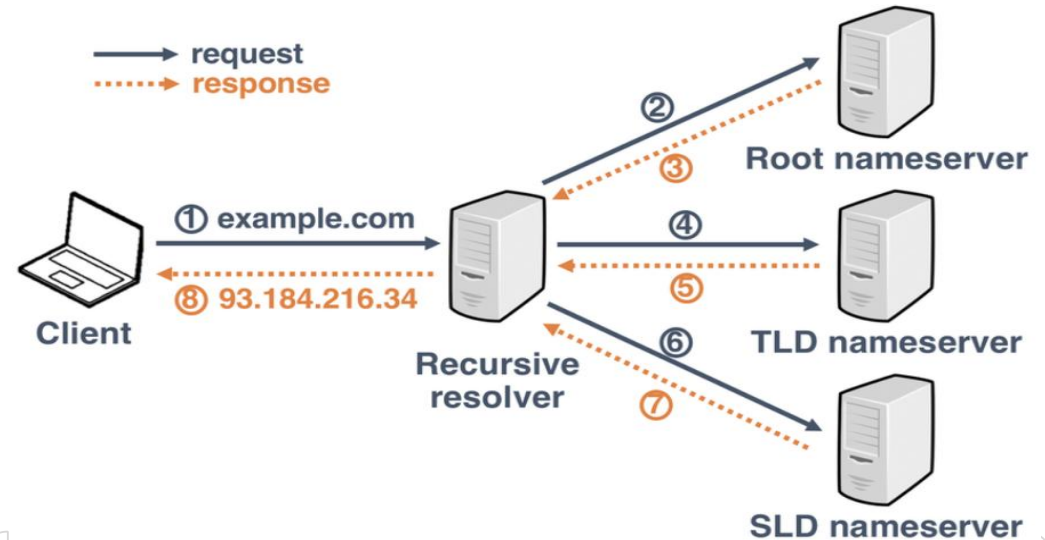
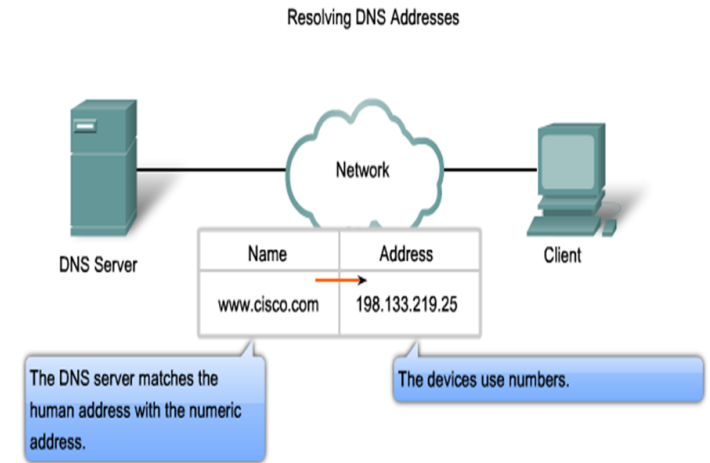
- A way to translate human-readable names into IP addresses

- How the client get the website

- 1- check the cash
- 2- check the hosts file
- 3- Ask DNS server

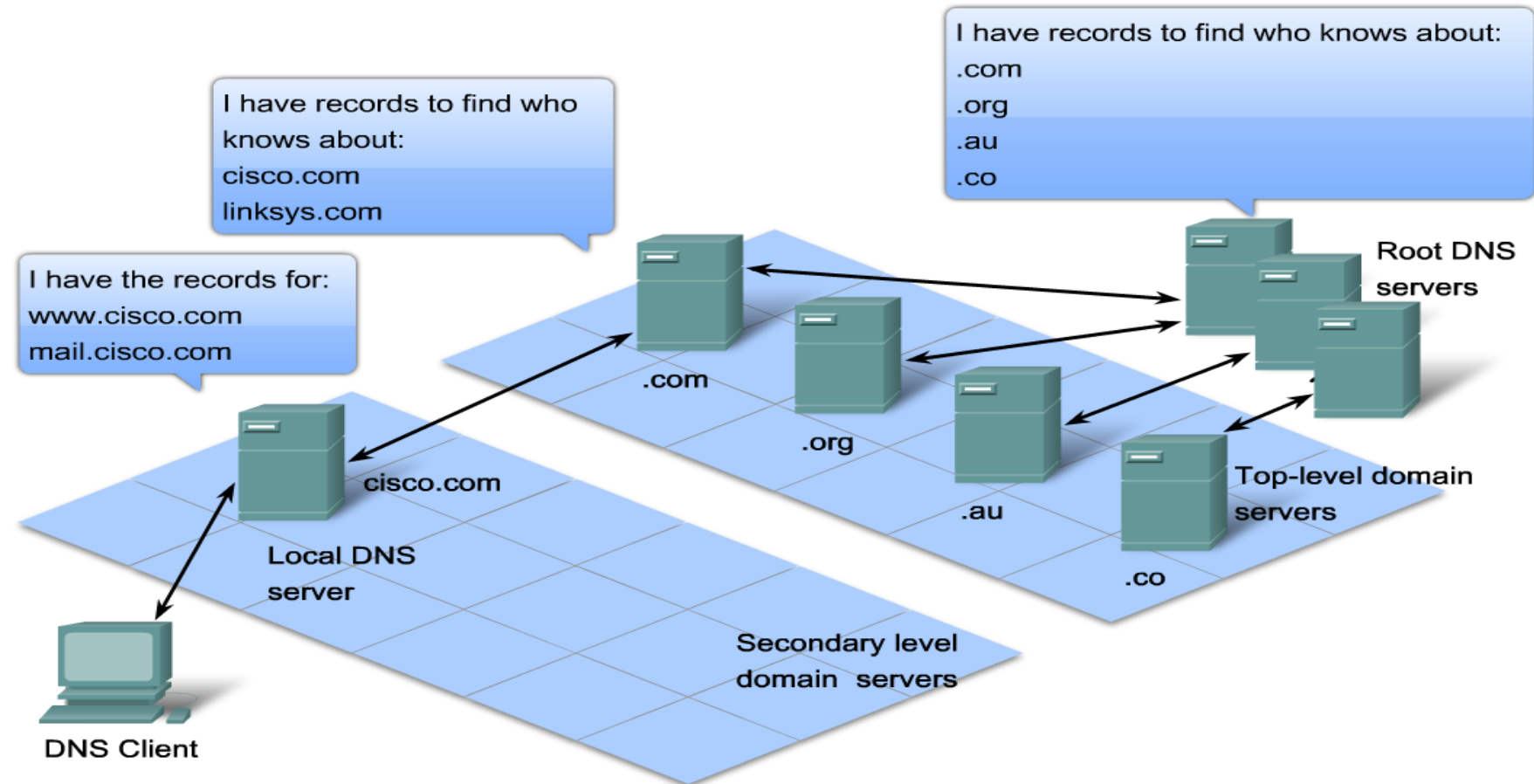
- List of Top Level Domains (TLDs)

Domain Name	Assigned To
com	Commercial organization
edu	Educational institution
gov	Government organization
mil	Military group
net	Major network support center
org	Organization other than those above
country code	A country



Part 1 (TCP/IP Protocol Architecture)

DNS Query

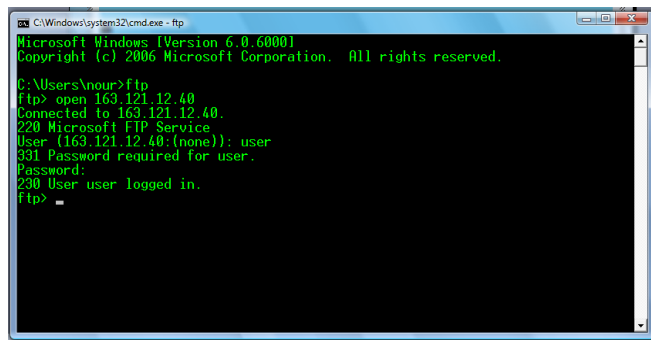


A hierarchy of DNS servers contains the resource records that match names with addresses.

Session 1 (TCP/IP Protocol Architecture)

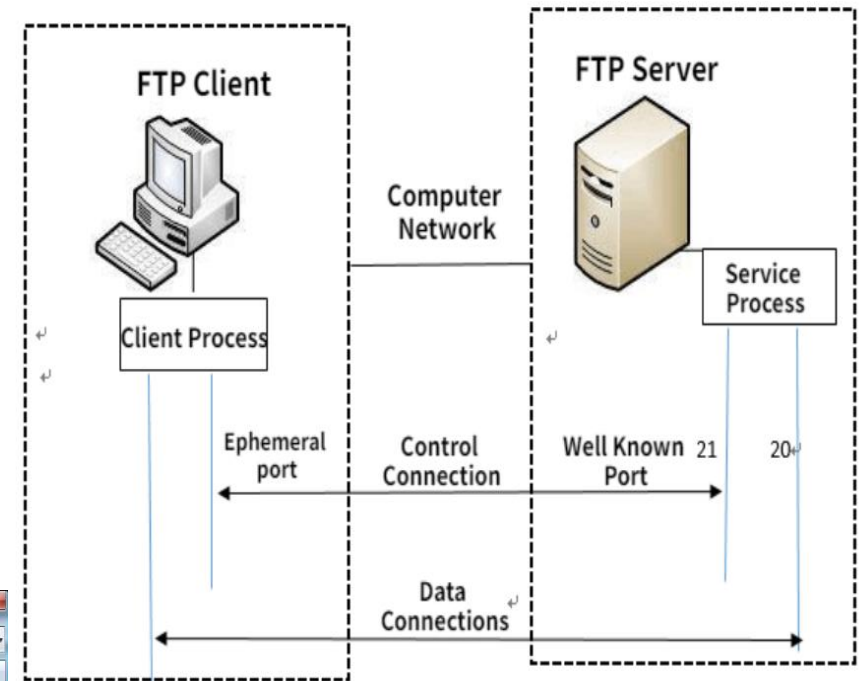
• FTP (File Transfer Protocol)

- a transmission protocol that provides reliable data transfer between hosts
- **FTP Client**
 - Use Internet Browser as FTP client.
 - Using MS Windows built-in FTP client
 - Third party programs “cute FTP”



```
C:\Windows\system32\cmd.exe - ftp
Microsoft Windows [Version 6.0.6000]
Copyright (c) 2006 Microsoft Corporation. All rights reserved.

C:\Users\snour>ftp
ftp> open 163.121.12.40
Connected to 163.121.12.40.
220 Microsoft FTP Service
User (163.121.12.40:(none)): user
331 Password required for user.
Password:
230 User user logged in.
ftp>
```



Working Principle of FTP

Session 1 (TCP/IP Protocol Architecture)



• Telnet /SSH or RDP

- Telnet/SSH is a user command and an underlying TCP/IP protocol for accessing remote computers.
- **Telnet/SSH**, an administrator can access someone else's computer remotely
- **Remote Desktop Protocol (RDP)** is a Microsoft proprietary protocol that enables remote connections to other computers,



Telnet VS SSH

SSH(Secure Shell)	Telnet
Runs on port 22	Runs on port 23
Very Secure Protocol	Not Secure Protocol
Only major protocol to access	Joint abbreviation
Difficult to decrypt	No data encryption
All popular Operating System	Linux , Windows



Session 1 (TCP/IP Protocol Architecture)

• Mail Server and Clients

• Mail Clients

- **Web based** : Hotmail ,gmail
- **Non web based** : Microsoft Outlook

• Mail Protocols

• SMTP (send mail transfer Protocol)

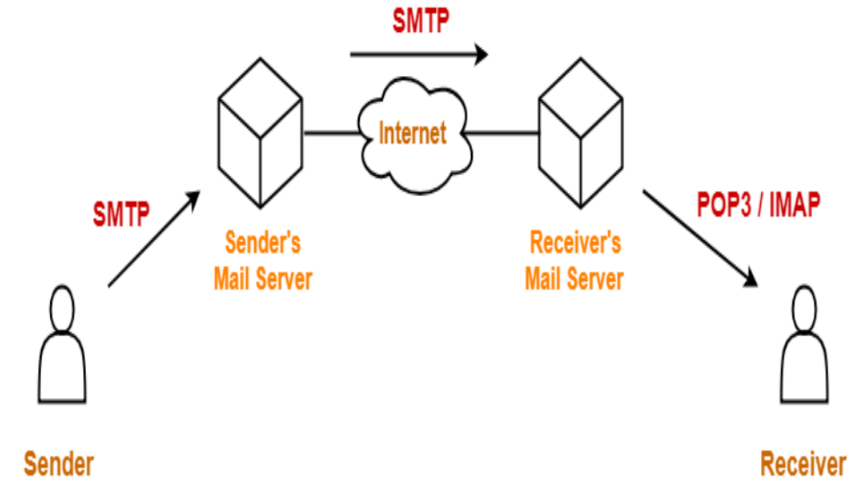
- send messages back and forth to other Mail Servers or Email Clients

• POP3 “Post Office Protocol version 3”

- the Email Client contacts the Mail Server to collect email messages
- Download messages on the hard disk
- can work Offline
- Keep the user's quota on the server

• IMAP4 “Internet Message Access Protocol version 4”

- Retrieve only message header



Session 1 Practices



- **Find your mac address**
 - Ipconfig /all
 - Get mac
- **Find your real IP addresses**
 - <https://www.whatismyip.com/>
- **Find your private IP addresses**
 - Ipconfig
 - Ipconfig /all
 - Ipconfig /release
 - Ipconfig /renew
 - ARP -a
- **Find current session and ports on your device**
 - Netstat -n
 - Netstat -a
- **Find The IP of the domain Yahoo.com**
 - Nslookup Yahoo.com



Thank You

